



Unit 4 · Outcome 2

Energy Sources

VCE Environmental Science · Units 3 & 4 · Full chapter revision

Every key-knowledge topic in this outcome, combined into one printable revision booklet with comparison tables, visual aids and worked answers.



Non-renewable energy sources

DEFINITION

What “non-renewable” actually means

A non-renewable energy source is one that exists in a finite stock and is replenished by natural processes far more slowly than humans consume it — effectively not at all on a human timescale. Once you burn this year’s coal, the Earth will not make more for tens of millions of years.

The contrast is with renewable sources (sun, wind, flowing water, tides, geothermal heat, biomass), which are replenished on a human timescale — the sun rises again tomorrow. The dividing line isn’t “will it run out one day” (the sun will too, in ~5 billion years); it’s the rate of natural replenishment versus the rate of human use.

◆ ANALOGY

🏠 Analogy — inheritance vs salary Treat the Earth’s non-renewables like a one-off inheritance sitting in a savings account: a huge balance, but no deposits are coming. Every dollar you spend lowers the total forever. Renewables are a salary — money that lands in the account every fortnight whether you spend it or not. A society running on non-renewables is living off savings; a society running on renewables is living within its income. The whole Energy outcome is really about shifting Australia from the first to the second.

Coal · oil · natural gas · coal seam gas · nuclear (uranium). Formed over millions of years; cannot be regrown within human time. Using them depletes the resource.

⚠ WATCH OUT

The trap hiding in this dot point Four of the five sources here are fossil fuels (coal, oil, natural gas, coal seam gas) — but the fifth, nuclear, is non-renewable yet NOT a fossil fuel. “Non-renewable” and “fossil fuel” are not synonyms. Keep that distinction primed; it’s tested almost every year.

D · SCRUB THE TIMELINE

Deep time: how fossil fuels formed

The four fossil fuels are, quite literally, ancient sunshine. Hundreds of millions of years ago, photosynthesis locked solar energy into the chemical bonds of living things. When that organic matter was buried before it could fully decay, then cooked under heat and pressure for geological ages, the carbon-rich remains became coal, oil and gas. Burning them today releases that stored sunlight — and the stored carbon.

Starts as dead vegetation in ancient swamps → compressed to peat → lignite (brown coal) → bituminous (black coal) → anthracite. More heat, time and pressure = higher carbon content and energy density. Victoria’s Latrobe Valley sits on enormous brown coal seams (younger, wetter, lower energy density).

Marine micro-organisms (plankton, algae) settle on ancient sea floors → buried in sediment → become a waxy substance called kerogen → moderate heat yields oil; greater heat “cracks” it into natural gas. The hydrocarbons migrate upward until trapped under impermeable rock — that’s a reservoir.

◆ KEY INSIGHT

✓ Carry-forward fact Because fossil fuels store carbon that was removed from the atmosphere millions of years ago, combusting them returns long-buried CO₂ to today's atmosphere. That single sentence is the bridge to KK03 (fossil fuels & the carbon cycle) and the whole Climate chapter.

DRAG TO SORT

The line students blur

◇ ACTIVE RECALL

Two confusions sink marks every year: (1) treating “non-renewable” and “fossil fuel” as the same thing (nuclear breaks that), and (2) thinking coal seam gas is a renewable or a totally different fuel (it's the same methane as natural gas, just unconventional). Sort these chips into the right bin to lock it in.

◆ KEY INSIGHT

✓ One-line distinctions to memorise Non-renewable $\supset$ fossil fuels + nuclear. • CSG vs natural gas: same gas (methane), different source — natural gas pools in a reservoir; CSG is held in coal seams and needs dewatering to release. • Nuclear isn't renewable just because it's low-carbon — uranium is a finite mined resource.

D · WATCH THE CHAIN

From fuel to the power point

Here's the thread that connects four of the five: coal, gas, CSG and even nuclear ultimately just make heat to boil water and spin a turbine. The fuel differs; the back end is almost identical. That's why this dot point sets up the energy-conversion and efficiency work in KK05–KK06.

◆ ANALOGY

🔗 Analogy — every thermal station is a kettle + pinwheel Picture a giant kettle (the boiler) with a pinwheel (the turbine) held in its steam. Whatever heats the kettle — burning coal, burning gas, or splitting uranium — the job is the same: make steam, spin the wheel, turn the generator. A gas turbine is the exception: hot combustion gases push the blades directly, no kettle required, which is why gas plants start so fast.

COMPARE TABLE

Advantages & disadvantages — the exam gold

VCAA almost always asks you to compare sources or justify a choice against criteria. Learn this grid as a mental scaffold: pick two or three rows that matter for the question and contrast them. Don't list everything — select the relevant criteria.

Criterion	Coal	Oil	Natural gas	Coal seam gas	Nuclear
Renewable?	No	No	No	No	No
Fossil fuel?	Yes	Yes	Yes	Yes	No
Main use	Electricity	Transport	Electricity, heat, LNG	Electricity, LNG	Electricity
Energy density	Moderate	High	Moderate	Moderate	Very high
Operating CO ₂	Highest	High	~½ of coal	~½ of coal*	Near-zero
Dispatchable?	Yes (baseload)	Yes	Yes (fast peaker)	Yes	Yes (baseload)
Key waste/ impact	Ash, mine, air pollution	Spills, emissions	Fugitive methane	Groundwater, salt, fracking	Radioactive waste
Status in Australia	Declining	Mostly imported	Major; transitional	Major (QLD)	Prohibited

* CSG's combustion emissions match natural gas; its overall footprint depends heavily on fugitive-methane leakage during extraction.

REAL FIGURES

The Australian & Victorian picture (2025–26)

Examiners reward students who anchor general claims in real data. Here are current, defensible numbers — use them, but quote the trend as much as the exact figure.

of Australia's 2025 electricity from fossil fuels — coal 42.7%, gas 16.2%, oil 1.7%. Renewables made up the other 39.5%.

of Victoria's electricity still comes from brown coal (Latrobe Valley) — the most CO₂-intensive fuel, now being phased out.

nuclear generation — prohibited under the EPBC Act 1999 & ARPANS Act 1998, despite Australia exporting uranium.

Three brown-coal stations remain, all with announced closure dates: Yallourn (EnergyAustralia) closing mid-2028, Loy Yang A (AGL) by 2035, and Loy Yang B (Alinta) intending to run until ~ 2047. Each closure is paired with new storage (e.g. big batteries) to firm renewables — a concrete example of the energy transition this whole outcome examines.

Coal: brown in VIC, black in NSW/QLD. Oil & conventional gas: Bass Strait, Cooper Basin (Moomba), North West Shelf — declining domestically. Coal seam gas: mainly Queensland's Surat/Bowen basins feeding Gladstone LNG, with reserves (and moratoria debates) in NSW & VIC. Nuclear: uranium mined in SA & NT for export only.

EXAM CRAFT

Decode the command word first

◆ COMMAND WORDS

Before writing a word, identify what the verb is asking for. The same content earns very different marks depending on whether you were told to describe or to evaluate. State it — no explanation. “Name two non-renewable energy sources.” → Coal and natural gas. Give the features/characteristics. What it is and how it works — no judgement. State the difference between two things, ideally on the same criterion (e.g. source of CSG vs natural gas). Give both similarities and differences, point-against-point — not two separate paragraphs. Give reasons / a causal chain. “Why does gas emit less CO₂ than coal?” Weigh advantages against disadvantages against stated criteria, then reach a supported judgement.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

MODEL ANSWER BUILDER

Writing like a top student — the P·E·L move



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

For any “explain / compare / justify” item, structure each idea as P·E·L — Point, Evidence, Link. Make the claim, back it with a specific fact or figure, then link it explicitly to the question. Walk the tabs below to see a 3-mark answer assembled.

AVOID THESE

Six traps that quietly cost marks

⚠ EXAM TRAPS

✗ “Nuclear is a fossil fuel.” ✓ Nuclear is non-renewable but NOT a fossil fuel — it uses fission of mined uranium, not combustion of buried organic matter. ✗ “Coal is renewable because we can keep mining it.” ✓ Renewability is about the rate of natural replenishment, not whether stock remains. Coal reforms over millions of years — far slower than we use it. ✗ “CSG is a different fuel from natural gas.” ✓ Same fuel (methane) — only the source differs: a reservoir (natural gas) vs adsorbed in coal seams (CSG). ✗ “Natural gas is clean / green / renewable.” ✓ It’s a lower-emission fossil fuel, still finite and still emitting CO₂ (plus methane if it leaks). “Cleaner than coal” ≠ “clean”. ✗ “Nuclear has no environmental downside because it emits no CO₂.” ✓ Its key disadvantage is long-lived radioactive waste (plus cost, build time and uranium being finite). ✗ Listing every advantage you know. ✓ Select criteria relevant to the question and compare point-against-point. A wall of facts with no comparison scores poorly on “compare/evaluate”.

SELF-MARK · 24 MARKS

Practice questions

★ PRACTICE & SELF-MARK

VCAA-style items escalating from 2 to 6 marks. Attempt each on paper first, then reveal a weak response, an annotated strong response, and the marking guide. Award yourself marks honestly — the dock bottom-right tracks your total. “Runs out” misses the rate idea; solar is renewable — a careless naming error costs a mark, and they’ve omitted coal seam gas.

- 1 — definition conveys finite AND replenished slower than used / not on a human timescale.
- 2 — all five correct (oil, coal, natural gas, coal seam gas, nuclear). 1 mark if 3–4 correct; 0 if a renewable is included as a main error.

★ PRACTICE & SELF-MARK

Circular and vague. No mention of adsorption/reservoirs or dewatering; “pollution” isn’t specific to CSG.

- 1 — natural gas located in a reservoir / free gas.
- 1 — CSG adsorbed in coal seams / released by dewatering.
- 1 — valid CSG advantage. 1 — valid CSG-specific disadvantage (groundwater, saline water, fracking, fugitive methane).

★ PRACTICE & SELF-MARK

The CO₂ point isn’t the reason it’s non-renewable; part b states a fact but never explains the two classifications.

- 1 — identifies the “renewable” claim as the error. 1 — correct reasoning (finite uranium / replenishment rate).
- 1 — non-renewable justified (finite uranium). 1 — not-fossil justified (fission, not combustion of organic matter).

★ PRACTICE & SELF-MARK

No named energy forms, no turbine/generator step; part b lacks a specific consequence.

- a: 1 — heat boils water to steam; 1 — steam (kinetic) turns turbine; 1 — generator produces electrical energy. (Naming energy forms required for full marks.)
- b: 1 — combustion vs fission as the heat source; 1 — a valid matching consequence (CO₂/pollution vs radioactive waste).

★ PRACTICE & SELF-MARK

Right instinct, but it's a thin list. No figures, no point-against-point comparison, criteria not clearly applied, judgement asserted not justified. Maybe 3/8. Emissions: Both are non-renewable, but gas emits roughly half the CO₂ per unit electricity of the closing brown coal, while nuclear produces near-zero operational CO₂. On emissions alone, nuclear is superior. Build time & transition fit: A gas peaker can be built and dispatched in a few years and starts within minutes, ideal for firming variable renewables now; a nuclear plant takes well over a decade, so it could not provide dispatchable supply during this transition. Waste & cost: Gas avoids long-lived waste but locks in ongoing CO₂ and fugitive-methane risk; nuclear avoids CO₂ but creates long-lived radioactive waste at high cost. Legality: Decisively, nuclear is prohibited in Australia (EPBC Act 1999; ARPANS Act 1998), so it is not currently a real option. Judgement: For reliability during the transition, the gas peaker is the better non-renewable option: it is fast to deploy, dispatchable and legal, and roughly halves emissions versus coal — even though nuclear scores better on operational emissions, its decade-plus build time and current prohibition make it unable to meet the transitional need.

- 1–2 — both correctly identified as non-renewable; relevant criteria selected.
- 3–5 — genuine point-against-point comparison on ≥3 criteria with specific evidence (CO₂ ~½ coal, build times, waste type, legality).
- 6–7 — emissions vs reliability tension handled; transition context applied.
- 8 — clear, supported judgement that follows from the comparison.

REVISE FAST

One-screen cheat sheet

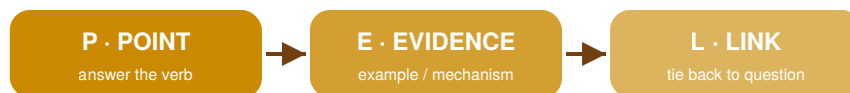
- Finite stock; replenished far slower than used → not on a human timescale.
- Renewable = replenished in-time (sun, wind, water, geothermal, biomass).
- Oil · Coal · Natural gas · Coal seam gas · Nuclear.
- First four = fossil fuels; nuclear = non-fossil non-renewable.
- Coal = buried plants (peat→lignite→bituminous→anthracite).
- Oil & gas = buried plankton → kerogen → oil (then gas with more heat).
- CSG = methane adsorbed in coal seams (dewater to release).
- Nuclear = fission of finite mined uranium.
- Coal & nuclear: heat → steam → turbine → generator.
- Gas: gas turbine (fast peaker); oil: mostly transport fuel.
- Coal: cheap, reliable / highest CO₂.

- Gas: ~½ coal's CO₂, dispatchable / still fossil, methane leaks.
- Nuclear: near-zero CO₂ / radioactive waste, slow, banned in AUS.
- Fossil 60.5% (coal 42.7, gas 16.2, oil 1.7); renewables 39.5%.
- VIC ~56% brown coal; Yallourn 2028, Loy Yang A 2035.
- Nuclear prohibited (EPBC 1999; ARPANS 1998).

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

point Gas is better than coal. (Vague — “better” how? No criterion.)

✓ STRONG / MODEL ANSWER

evidence Burning methane releases roughly half the CO₂ per unit of electricity that brown coal does, and gas turbines can start within minutes, so they can firm variable solar and wind when output drops.

✗ WEAK ANSWER

evidence It's cleaner and pretty quick. (No figure, no mechanism — examiner can't award the mark.)

✓ STRONG / MODEL ANSWER

link This lets gas bridge the period between a coal-dominated grid and a fully renewable one — reducing emissions now without sacrificing reliability — which is why it's labelled “transitional” rather than a long-term solution.

✗ WEAK ANSWER

link So that's why people use gas. (Doesn't reconnect to “transitional”.)

✓ STRONG / MODEL ANSWER

Full P-E-L response Natural gas is called a “transitional” fossil fuel because it emits far less CO₂ than coal yet remains dispatchable. Burning methane releases roughly half the CO₂ per unit of electricity of brown coal, and fast-starting gas turbines can firm variable solar and wind within minutes. This lets gas bridge the shift from a coal-heavy grid to a renewable one — cutting emissions now without risking reliability — so it is treated as a stepping stone rather than a permanent, sustainable solution.

The seven renewables, and why they never run out

What makes a source "renewable"?

The whole study design splits energy into non-renewable (KK01) and renewable (KK02). The line between them is not "does it pollute" or "is it clean" — it is about rate.

A renewable energy source is replenished by natural processes at a rate equal to or faster than the rate at which it is used, so it is not depleted on a human timescale.

Non-renewable sources (coal, oil, gas, uranium) formed over millions of years and are consumed far faster than they re-form — so the stock shrinks. Renewables ride an ongoing flow (sunlight, wind, the water cycle, the tides, Earth's internal heat, this season's plant growth) that keeps arriving.

◆ NOTE

Analogy · stock vs flow Non-renewables are a savings account you can only withdraw from — every kWh spent is gone. Renewables are a salary that lands every month: spend within the inflow and you never go broke. Coal is the inheritance; sunshine is the paycheck.

◆ NOTE

Trap · renewable ≠ zero-emission A source can be renewable but still emit greenhouse gases (burning biomass releases CO₂), and a renewable can carry big environmental costs (a hydro dam floods habitat).

"Renewable" describes replenishment rate, not cleanliness. Keep the two ideas separate in exam answers.

The seven sources, one card each

For every source, learn the same four things: the natural flow it taps, the energy-conversion chain to electricity, why it's renewable, and its standout advantage / disadvantage. The conversion chains matter — they connect straight to KK05.

◆ NOTE

Memory hook "Sun-Wind-Water-Tide, Heat-and-Harvest" — solar & wind (intermittent), hydro & tidal (water-driven), geothermal (Earth's heat), then the two harvest sources: biomass (solid) and biofuels (liquid). Seven sources, three families.

:00 · CLEAR & BREEZY * CLEAR/BREEZY ☀️ CLOUDY/STILL 🌪️ WINDY STORM 3D NOT AVAILABLE — USE THE READOUT PANEL BELOW; IT SHOWS THE SAME LIVE OUTPUT VALUES FOR EACH SOURCE.

06 3D · The intermittency lab

Renewables don't all behave the same across a day. Drag the time-of-day slider and flip the weather — watch which sources surge, fade or stay rock-steady. This is the single most examined idea in the chapter, and it sets up base/peak load (KK09).

SOLID · BURNED DIRECTLY

Biomass vs biofuels — the classic mix-up

The study design lists these as two separate sources, and examiners love testing whether you can tell them apart. Both come from recently living matter; the difference is form and processing.

Organic matter burned to release stored chemical energy as heat. Wood, crop stubble, bagasse (sugarcane fibre), forestry residue, even municipal organic waste. Burned to raise steam → turbine → generator.

Mirran Plains example: a small bagasse/stubble boiler at the edge of the cane and wheat district burns harvest residue that would otherwise rot.

Liquid or gaseous fuels processed from biomass to power engines — ethanol (from sugar/starch, blended as E10) and biodiesel (from canola or used cooking oil). Biogas (methane) from anaerobic digestion also counts.

Mirran Plains example: local canola → biodiesel runs the farm machinery and a backup generator; the servo sells E10.

◆ NOTE

The "carbon-neutral in principle" idea Plants absorb CO₂ as they grow, then release roughly that same CO₂ when burned — so the combustion carbon can be close to balanced if new crops regrow to reabsorb it. That's why biomass/biofuels are called approximately carbon-neutral. The catch examiners reward: it's only near-neutral once you subtract the fossil energy used to grow, harvest, process and transport the feedstock — and not if forests are cleared faster than they regrow.

Intermittent vs dispatchable — the deciding axis

Beyond "renewable or not", the property that drives real planning decisions is whether a source can deliver power on demand. This single table answers most comparison questions in the chapter.

Source	Natural flow	Renewable?	On-demand (dispatchable)?	Greenhouse at point of use
Solar PV	Sunlight	Yes	No — intermittent (day/cloud)	~Zero
Wind	Moving air	Yes	No — intermittent (wind speed)	~Zero
Hydro-electric	Falling/stored water	Yes	Yes — fast, storable	~Zero*
Tidal	Ocean tides	Yes	Predictable but cyclic	~Zero
Geothermal	Earth's internal heat	Yes	Yes — steady baseload	Low
Biomass	This season's plant growth	Yes	Yes — fuel can be stored	CO ₂ released (≈reabsorbed)
Biofuels	Crops → liquid fuel	Yes	Yes — fuel stored	CO ₂ released (≈reabsorbed)

*Reservoir hydro can emit methane from flooded vegetation, especially in warm climates — a nuance worth a mark if a question pushes on it.

◆ NOTE

Why this matters Solar and wind are cheap and clean but intermittent — they generate when nature decides, not when we flick the switch. Hydro (especially pumped hydro like Snowy 2.0), geothermal, biomass and stored biofuels are dispatchable — they can fill the gaps. A good energy mix pairs cheap intermittent generation with dispatchable backup and storage. Hold this thought for KK09.

Command-word decoder

◆ COMMAND WORDS

Match what the verb is asking for. Tap a word to expand.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

P·E·L writing trainer



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

The house writing frame: Point → Explain → Link. Draft a response to the prompt below; the trainer lights up as you hit the ideas examiners reward. Prompt: Explain why solar and wind are described as intermittent, and why this makes a dispatchable renewable like hydro-electricity valuable in the same grid. (4 marks) State that solar/wind depend on a natural condition that varies, so output is not constant. Link the variation to the source (no sun at night/cloud; no output when wind is too low/high) and contrast with water that can be stored and released on demand. Conclude that dispatchable hydro fills the gaps, keeping supply matched to demand — directly answering "why valuable".

Good vs bad answer, side by side

Question: Compare a solar farm and a geothermal plant as sources of electricity for the Mirran Plains zone, referring to renewability and reliability of supply. (4 marks)

• **NOTE**

Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored, and the geothermal point is dropped. "Compare" needs both sources weighed against the same criteria.

• **NOTE**

Both sources weighed on the same two criteria, the intermittent/dispatchable contrast is explicit, and it lands a judgement tied to the scenario. Full marks.

Practice exam questions

★ **PRACTICE & SELF-MARK**

VCAA-style questions totalling 20 marks. Write your answer, reveal the model and marking guide, then self-mark with the stars. Your progress saves automatically.

One-screen cheat sheet

♦ **RECAP / CHEAT SHEET**

Replenished by natural processes at a rate $\geq$ the rate of use, so not depleted on a human timescale. About rate, not cleanliness. Solar, wind, hydro-electric, tidal, geothermal, biomass, biofuels. Three families: sun/wind, water (hydro/tidal), then Earth-heat + harvest. Solar & wind — generate when nature allows, not on demand. Output varies with sun/cloud and wind speed. Hydro, geothermal, biomass, stored biofuels — power on demand. Tidal is predictable but cyclic. Biomass = solid matter burned directly. Biofuel = liquid/gas fuel refined from biomass (ethanol, biodiesel, biogas). Bio sources release CO_2 but regrowth reabsorbs $\approx$ the same — near-neutral, minus fossil energy used to grow/process/transport. Wind/hydro/tidal: kinetic $\rightarrow$ electrical. Solar PV: radiant $\rightarrow$ electrical (no turbine). Geo/biomass/biofuel: thermal $\rightarrow$ kinetic $\rightarrow$ electrical. Snowy 2.0 pumped hydro (2.2 GW, dispatchable storage); Golden Plains wind (VIC); large solar farms (Western Downs, New England); geothermal direct-use; ethanol E10 & canola biodiesel.

EXAM TECHNIQUE

★ **Worked answers — weak vs strong**

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

Solar is good because it's clean and free from the sun. Geothermal uses heat from the ground which is also good. Both are renewable so they're better than coal. The Mirran Plains zone should use solar because Australia is sunny. Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored, and the geothermal point is dropped. "Compare" needs both sources weighed against the same criteria.

✓ STRONG / MODEL ANSWER

Both are renewable: sunlight and Earth's internal heat are replenished far faster than they are used. They differ on reliability of supply. Solar PV is intermittent — it generates only in daylight and drops under cloud, so its output varies and cannot be guaranteed at peak evening demand. Geothermal taps a continuous heat flow, so it runs steadily day and night as dispatchable baseload. For Mirran Plains, solar offers cheap bulk daytime energy, while geothermal provides the reliable around-the-clock supply solar lacks — so the two are complementary rather than competing. Both sources weighed on the same two criteria, the intermittent/dispatchable contrast is explicit,

Fossil fuels & the carbon cycle

What the dot point is really asking

The VCAA key knowledge statement is just nine words. Examiners hide a lot inside them.

"The consequences of fossil fuel combustion for the carbon cycle."

The chemical reaction — hydrocarbons + oxygen → carbon dioxide + water + energy. You must be able to write and read it.

The movement of carbon between stores (atmosphere, oceans, biosphere, lithosphere) via fluxes like photosynthesis, respiration and combustion.

What changes when we burn fossil fuels: where carbon moves, how fast, and why natural sinks can't keep up.

◆ KEY INSIGHT

⊙ The one-sentence answer Fossil fuel combustion takes carbon that was locked in the lithosphere for hundreds of millions of years and transfers it — very rapidly — into the atmosphere as CO₂, adding carbon to the fast carbon cycle faster than natural sinks can remove it, so atmospheric CO₂ rises. Everything below unpacks that sentence.

The combustion reaction

Carbon doesn't appear from nowhere when fuel burns. Combustion is just rapid oxidation — atoms rearranging, not vanishing.

A fossil fuel is essentially stored chemical potential energy held in the bonds between carbon and hydrogen atoms. When the fuel reacts with oxygen, those bonds break and reform as CO₂ and H₂O, releasing the energy as heat. Complete combustion of a generic hydrocarbon looks like this:

◆ NOTE

⚖ Conservation of mass — the heart of the carbon cycle Every carbon atom on the left of the arrow appears on the right as part of a CO₂ molecule. Combustion can't destroy carbon; it can only relocate it — out of the solid fuel and into the gaseous atmosphere. That relocation is the entire story of this dot point.

The fuel matters because different fossil fuels carry different amounts of carbon per unit of energy. Coal is almost pure carbon, so it releases the most CO₂ per joule. Natural gas (methane) has four hydrogens for every carbon, so a larger share of its energy comes from forming water rather than CO₂ — it is the "cleanest" fossil fuel per unit of energy, though still a net source.

⚠ WATCH OUT

Incomplete combustion (worth knowing, not the focus) With limited oxygen you also get carbon monoxide (CO) and soot (carbon particulates): $2C + O_2 \rightarrow 2CO$. These are pollution/health issues, but for the carbon cycle dot point the examinable point is still that the fuel's carbon ends up in the atmosphere.

Where the carbon came from

To understand the consequence, you first have to know that this carbon is not new — it's a withdrawal from a very old account.

Coal, oil and gas are called fossil fuels because they are the buried, compressed remains of living things. Hundreds of millions of years ago, plants and plankton used photosynthesis to pull CO_2 out of the atmosphere and build carbon-rich tissue. When they died, some of that carbon was buried before it could decompose, and over geological time — heat, pressure, millions of years — it became coal seams and oil and gas reservoirs deep in the lithosphere.

So the carbon in the Kalduke Valley coal seam was removed from the atmosphere ~300 million years ago and has been locked out of circulation ever since. It is part of the long-term (slow) carbon cycle — a store that, left alone, would only return to the atmosphere over thousands to millions of years through slow geological processes.

◆ ANALOGY

Analogy · the carbon bank vault Imagine carbon as money. The atmosphere, plants and surface ocean are the cash in circulation — constantly changing hands (the fast cycle). Fossil fuels are gold bars locked in a deep vault, sealed for 300 million years. Combustion is cracking the vault open and dumping all that long-stored wealth into circulation at once. The total money in the world goes up, and you get inflation — here, the "inflation" is rising atmospheric CO_2 .

The fast cycle, the slow cycle, and the shortcut

The carbon cycle runs on two clocks. Fossil fuel combustion is a shortcut that wires the slow clock straight into the fast one.

The fast carbon cycle moves carbon between the atmosphere, living things, soils and the surface ocean over years to decades — photosynthesis pulls it down, respiration and decomposition push it back up. It's roughly balanced: over a year, sinks absorb about as much as sources release.

The slow carbon cycle moves carbon into and out of rocks, sediments and fossil fuels over thousands to millions of years — weathering, sedimentation, volcanism. Burning fossil fuels takes carbon out of that slow store and injects it into the fast cycle almost instantly. Use the scene below to see the difference.

◆ KEY INSIGHT

⊙ The key idea Combustion is a one-way shortcut from the slow cycle into the fast cycle. It doesn't add a new "drain" to balance it — it just opens a new tap. Carbon that took 300 million years to lock away comes back in a single human lifetime.

The source–sink ledger

Atmospheric CO_2 is a balance between what flows in (sources) and what flows out (sinks). Tip the balance and the level changes.

◆ ANALOGY

Analogy · the bathtub Picture the atmosphere as a bathtub of CO_2 . The taps are sources (respiration, decomposition, volcanism, and now combustion). The drain is the sinks (photosynthesis, dissolving into the ocean). For most of Earth's recent history the taps and drain matched, so the water level (CO_2) held steady. Fossil fuels open an extra tap the drain was never sized for — so the level rises year on year.

Crucially, the natural sinks do absorb some of our emissions — the ocean and land biosphere together take up roughly half of the CO_2 we emit. But they can't take all of it, so the remainder accumulates. Drag the combustion lever and watch the tub.

◆ NOTE

■ The numbers examiners love Pre-industrial atmospheric CO_2 was about 280 ppm. It has now passed 430 ppm (Mauna Loa, 2025–26) — roughly 50% higher than before the Industrial Revolution, and higher than at any time in over 2 million years. The rate of rise (~ 2.5 ppm/yr this decade) is around 100 times faster than natural increases at the end of the last ice age.

The consequences cascade

"Consequences for the carbon cycle" branches into three examinable outcomes. Learn them as a chain.

Sources exceed sinks, so CO_2 accumulates in the atmosphere — the upward Keeling-curve trend. This is the primary, direct carbon-cycle consequence.

More CO_2 absorbs more outgoing infrared radiation, warming the lower atmosphere — the link into the Climate Change outcome (U4.O1).

The ocean absorbs much of the extra CO_2 . That shifts a different part of the carbon cycle — and changes ocean chemistry.

The third branch is the one students forget, so it's worth seeing. When CO_2 dissolves in seawater it doesn't just disappear — it reacts with water to form carbonic acid, which releases hydrogen ions and lowers the ocean's pH:

Those extra hydrogen ions make it harder for corals, shellfish and plankton to build calcium-carbonate shells and skeletons. The ocean has already absorbed enough CO_2 to lower its surface pH by about 0.1 units since pre-industrial times — roughly a 30% increase in acidity. Watch the scene: as you raise atmospheric CO_2 , more dissolves, pH drops, and the shell struggles to form.

⚠ WATCH OUT

Stay on the dot point In a KK03 answer, ocean acidification matters because it shows fossil-fuel carbon entering a different store of the carbon cycle (the hydrosphere) and changing it. If a question only asks about the carbon cycle, don't drift into a full climate-change essay — make the carbon-store link explicit.

The distinction that wins marks: fossil vs biofuel carbon

Burning wood also releases CO_2 . So why is burning coal worse for the carbon cycle? This is the single most rewarded idea in the dot point.

Was removed from the atmosphere ~300 million years ago and locked in the slow cycle. Burning it adds carbon that was not part of the active cycle — a net injection. Atmospheric CO₂ rises. Not carbon-neutral.

Was removed from the atmosphere recently (months–decades) by the growing plant. Burning it returns carbon already in the fast cycle. If regrown, the loop closes — broadly carbon-neutral over short timescales.

◆ KEY INSIGHT

◎ Say this in the exam It's not that fossil fuels release "more" CO₂ per breath — it's that they introduce "new" carbon into the fast cycle from a long-term geological store, whereas biofuels merely recycle carbon the plant recently absorbed. Same gas, very different effect on the cycle's balance.

Try it. Drag each statement into the correct bin. (On touch, tap a chip then tap a bin.)

Australian context: the Kalduke Valley & brown coal

Victoria's electricity has long leaned on the most carbon-intensive fossil fuel there is — a perfect KK03 case study.

Our fictional Wirndook Power Station in the Kalduke Valley burns brown coal (lignite), mirroring Victoria's real Latrobe Valley generators. Brown coal is the worst case for this dot point: it is high in moisture and low in energy density, so a power station must burn more of it to produce each unit of electricity — releasing more CO₂ per kilowatt-hour than black coal or gas. Every tonne burned is a withdrawal from a Gippsland coal seam that has held its carbon for tens of millions of years.

Highest CO₂ per unit energy of the common fuels — the carbon-cycle "worst offender". Dominant in Victoria's legacy grid.

Higher energy density, somewhat less CO₂ per kWh than brown coal. Common in NSW/Queensland generation and exports.

Lowest-carbon fossil fuel per unit energy, but still a net source — it still moves slow-cycle carbon into the air.

◆ NOTE

Real anchors you can name Victoria's Latrobe Valley brown-coal stations (e.g. the closure of Hazelwood in 2017) and Australia's position as one of the world's largest coal and LNG exporters mean much of the carbon we dig up is burned overseas — but the atmosphere is global, so it still loads the same shared fast cycle.

Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you how much to write. Tap a card to see what the examiner wants.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

Six exam traps

⚠ EXAM TRAPS

These are the mistakes that quietly cost marks on this dot point every year.

The P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

Environmental Science extended responses reward structure. P·E·L turns a vague "explain" into a mark-grabbing paragraph. State your claim in the question's own language. "Fossil fuel combustion transfers stored carbon into the atmosphere." Give the mechanism + a fact or figure. "The carbon was locked in the lithosphere for ~300 My; burning it adds CO₂ faster than sinks absorb it." Tie back to the carbon cycle / the command word. "...so atmospheric CO₂ rises — a net shift from the slow cycle into the fast cycle."

◆ NOTE

✓ Why this scores full marks It names the store and the timescale, includes the reaction, explains the source–sink imbalance with a figure, and finishes with the net carbon-cycle consequence — four distinct, markable ideas, tightly linked.

Practice questions

★ PRACTICE & SELF-MARK

Five VCAA-style questions, escalating from 2 to 6 marks (20 total). Open each, attempt it, then compare a weak and a strong response — and mark your own. "A fossil fuel is something like coal or oil that you burn for energy. The carbon comes from old dead things."

• **NOTE**

Names examples but never defines the term, and "old dead things" doesn't identify the process. The question wants photosynthesis. Likely 0–1 / 2.

★ **PRACTICE & SELF-MARK**

"A fossil fuel is a non-renewable energy source formed from the buried, compressed remains of ancient organisms over millions of years. Its carbon was originally removed from the atmosphere by photosynthesis in those plants/plankton."

- 1 mark — fossil fuel defined as a non-renewable fuel formed from remains of ancient organisms over geological time.
- 1 mark — carbon removed from the atmosphere by photosynthesis.

★ **PRACTICE & SELF-MARK**

"When you burn coal it makes smoke and CO_2 and that goes into the air. Carbon + oxygen = energy. "

• **NOTE**

The equation is wrong (omits CO_2 as the product) and there's no mention of which systems carbon moves between. Vague. 1 / 3.

★ **PRACTICE & SELF-MARK**

"Combustion: hydrocarbon + $\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{energy}$ (for coal, $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$). This takes carbon stored in the lithosphere (the solid fuel) and releases it as carbon dioxide into the atmosphere, transferring carbon from a geological store into the air."

• **NOTE**

Correct equation + explicit lithosphere→atmosphere transfer. 3 / 3.

- 1 mark — correct combustion equation showing CO_2 as a product.
- 1 mark — identifies carbon source as the fuel / lithosphere (geological store).
- 1 mark — identifies carbon released to the atmosphere as CO_2 .

★ **PRACTICE & SELF-MARK**

"Burning fossil fuels makes lots of CO_2 and that's bad for the environment and causes global warming. The trees can't absorb it all because there's too much pollution."

• **NOTE**

Drifts into climate impacts, never explains the source–sink imbalance or where the carbon came from. "Too much pollution" isn't a mechanism. 1 / 4.

★ PRACTICE & SELF-MARK

"Fossil fuels hold carbon that was removed from the atmosphere hundreds of millions of years ago and stored in the lithosphere — part of the slow carbon cycle. Burning them injects this carbon into the fast cycle as CO₂ very rapidly. Natural sinks (ocean, biosphere) absorb roughly half of it, but the rate of release now exceeds the rate of uptake, so the surplus accumulates and atmospheric CO₂ rises — from ~280 ppm pre-industrial to over 430 ppm today."

• NOTE

Origin + slow/fast framing + explicit imbalance + figures. 4 / 4.

- 1 mark — fossil carbon was stored long-term in the lithosphere (slow cycle).
- 1 mark — combustion releases it rapidly into the atmosphere/fast cycle.
- 1 mark — sinks still operate but the rate of release exceeds rate of uptake (imbalance).
- 1 mark — therefore net accumulation; CO₂ rises (figure/data rewarded).

★ PRACTICE & SELF-MARK

"Wood is renewable and coal is non-renewable, so wood is better. They both make CO₂ but wood is cleaner and more natural."

• NOTE

Renewable vs non-renewable is true but doesn't address the carbon cycle. No mention of where each fuel's carbon originated or the carbon-neutral idea. Only one side compared. 1–2 / 5.

★ PRACTICE & SELF-MARK

" Similarity: both release CO₂ into the atmosphere when burned ($C + O_2 \rightarrow CO_2$). Difference — origin of carbon: brown coal's carbon was locked in the lithosphere for tens of millions of years (slow cycle), so burning it adds 'new' carbon to the fast cycle, raising atmospheric CO₂. Wood's carbon was absorbed by the tree within recent decades, so burning it only returns carbon already in the fast cycle. Net effect: if the wood is replanted and regrows, it is broadly carbon-neutral over the short term, whereas brown coal is a net source that permanently increases the fast-cycle carbon pool."

• NOTE

Similarity + difference in origin + slow/fast framing + carbon-neutral vs net source. Both sides, throughout. 5 / 5.

- 1 mark — similarity: both release CO₂ on combustion.
- 1 mark — coal's carbon originated in the slow cycle / lithosphere (long timescale).
- 1 mark — biofuel's carbon was recently taken from the atmosphere (fast cycle).
- 1 mark — biofuel broadly carbon-neutral if regrown (closed loop).
- 1 mark — coal is a net source that increases the fast-cycle carbon pool.

★ PRACTICE & SELF-MARK

"Fossil fuels cause global warming which melts ice and raises sea levels. This is very bad. The ocean gets warmer and animals die. We should use solar instead."

• NOTE

All downstream climate impacts — no carbon-cycle framing, no acidification chemistry, no judgement that actually engages "evaluate". Off the dot point. 1–2 / 6.

★ PRACTICE & SELF-MARK

"The statement is well supported. Direct carbon-cycle consequence: combustion transfers slow-cycle carbon (lithosphere) into the atmosphere as CO_2 , and because release exceeds sink uptake, atmospheric CO_2 has risen $\sim 280 \rightarrow 430$ ppm. Ocean consequence: the ocean absorbs much of this surplus — a transfer of carbon into the hydrosphere — but the dissolved CO_2 reacts with water ($\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$), releasing H^+ ions and lowering ocean pH (~ 0.1 units, $\approx 30\%$ more acidic). This impairs calcifying organisms (corals, shellfish). Judgement: so even setting warming aside, combustion measurably alters two carbon stores — atmosphere and ocean — confirming the consequences for the carbon cycle are broad, not limited to surface temperature."

• NOTE

Carbon-cycle framing + acidification chemistry + figures + a clear evaluative judgement that answers the verb. 6 / 6.

- 1 mark — combustion moves carbon from lithosphere (slow) to atmosphere (fast) as CO_2 .
- 1 mark — source–sink imbalance $\rightarrow$ atmospheric CO_2 rises (data rewarded).
- 1 mark — ocean absorbs CO_2 — carbon transferred into the hydrosphere.
- 1 mark — acidification mechanism: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow$ carbonic acid $\rightarrow \text{H}^+ \rightarrow$ lower pH.
- 1 mark — consequence for marine calcifiers / a named figure (≈ 0.1 pH, $\approx 30\%$).
- 1 mark — supported evaluative judgement addressing "beyond warming".

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

A fossil fuel is something like coal or oil that you burn for energy. The carbon comes from old dead things." Names examples but never defines the term, and "old dead things" doesn't identify the process. The question wants photosynthesis. Likely 0–1 / 2.

✓ STRONG / MODEL ANSWER

A fossil fuel is a non-renewable energy source formed from the buried, compressed remains of ancient organisms over millions of years. Its carbon was originally removed from the atmosphere by photosynthesis in those plants/plankton." Definition + the named process. 2 / 2.

✗ WEAK ANSWER

When you burn coal it makes smoke and CO₂ and that goes into the air. Carbon + oxygen = energy. " The equation is wrong (omits CO₂ as the product) and there's no mention of which systems carbon moves between. Vague. 1 / 3.

✓ STRONG / MODEL ANSWER

Combustion: hydrocarbon + O₂ → CO₂ + H₂O + energy (for coal, C + O₂ → CO₂). This takes carbon stored in the lithosphere (the solid fuel) and releases it as carbon dioxide into the atmosphere, transferring carbon from a geological store into the air." Correct equation + explicit lithosphere→atmosphere transfer. 3 / 3.

✗ WEAK ANSWER

Burning fossil fuels makes lots of CO₂ and that's bad for the environment and causes global warming. The trees can't absorb it all because there's too much pollution." Drifts into climate impacts, never explains the source–sink imbalance or where the carbon came from. "Too much pollution" isn't a mechanism. 1 / 4.

✓ STRONG / MODEL ANSWER

Fossil fuels hold carbon that was removed from the atmosphere hundreds of millions of years ago and stored in the lithosphere — part of the slow carbon cycle. Burning them injects this carbon into the fast cycle as CO₂ very rapidly. Natural sinks (ocean, biosphere) absorb roughly half of it, but the rate of release now exceeds the rate of uptake, so the surplus accumulates and atmospheric CO₂ rises — from ~280 ppm pre-industrial to over 430 ppm today." Origin + slow/fast framing + explicit imbalance + figures. 4 / 4.

Fossil fuel use & peak oil

"Changes in the rate of fossil fuel use over time and the concept of peak oil."

A1 Three fossil-fuel eras

For almost all of human history, our energy came from muscle, wood, wind and water. Fossil fuels changed that — but they did not all arrive at once. They scaled up in sequence, each unlocked by a new technology.

From the Industrial Revolution (~1760s onward), coal powered steam engines, iron-making and railways. It was the first fossil fuel to scale, and it dominated global energy into the early 20th century.

After the first commercial well (1859) and the internal combustion engine, oil surged through the 20th century. Cars, trucks, ships, aviation and petrochemicals made oil the backbone of modern transport.

Gas grew strongly from the mid-1900s for heating, industry and electricity. It is the fastest-growing fossil fuel today, often framed as a lower-carbon "bridge" fuel — though it is still a fossil fuel.

Add them together and the global picture is a roughly exponential climb through the 1900s. But notice the shape carefully: total use is still rising, while the rate of growth is slowing, and coal use has plateaued (and may have peaked) in many regions as gas and renewables displace it. "The rate of fossil fuel use over time" is not a single number — it is a curve with a story.

A2 What pushes the rate up — and back down

If a question asks you to explain changes in the rate, you need causes, not just a description of the line. Examiners want the drivers on both sides.

- Population growth — more people, more energy demanded
- Industrialisation & economic growth — factories, construction, freight
- Rising living standards — appliances, cars, air-con, travel
- Transport & electrification spreading worldwide
- Cheap, energy-dense, easy to store — fossil fuels were simply convenient
- Energy efficiency — doing more with each unit of fuel
- Renewables & substitution — solar, wind and gas displacing coal
- Policy & regulation — carbon targets, emissions limits, EV uptake
- Price — high prices curb demand and reward alternatives
- Resource depletion — the easy reserves get used first

◆ ANALOGY

Analogy — a city's traffic The rate of fossil fuel use is like cars on a freeway. For a century, more lanes kept opening (population, industry, transport) and traffic only grew. Lately, working-from-home, trains and electric cars (efficiency, substitution, policy) have started easing the flow in wealthy nations — even though, globally, the freeway is still busy.

B1 Peak oil: Hubbert and the curve

Now the second half. "Peak oil" is a precise idea — and the single most-tested thing about it is what it does not mean.

The point in time at which the maximum rate of oil extraction (production) is reached, after which the rate of production enters a terminal, irreversible decline.

In 1956, geoscientist M. King Hubbert reasoned that because oil is a finite, non-renewable resource, its production over time must trace a roughly symmetrical bell-shaped curve — now called a Hubbert curve. Production rises as fields are found and developed, reaches a peak, then falls as reserves deplete. Hubbert predicted that conventional oil production in the United States would peak around 1970 — and it did, almost exactly.

⚠ WATCH OUT

The trap that costs the most marks Peak oil does NOT mean oil is "running out." At the peak, roughly half of the recoverable oil still remains underground. What changes is the rate you can pull it out. And because the cheapest, easiest, highest-quality oil is taken first, the remaining half is harder, slower and more expensive to extract.

💠 ANALOGY

Analogy — the lolly jar at a party Picture a big jar of lollies. Everyone grabs the easy ones off the top first, so the grab rate (lollies per minute) climbs fast. It peaks while the jar is still about half full — then slows, because people now have to dig awkwardly for the ones stuck at the bottom. Peak oil is the peak of the grab rate, not the empty jar. The lollies aren't gone; they're just harder to reach.

B2 Why the global peak keeps moving

Hubbert nailed the US peak, but global "peak oil" has been predicted — and missed — many times. Knowing why turns a 2-mark recall into a 4-mark explanation.

Factor	How it pushes the peak later (or reshapes the curve)
New discoveries	Fields once thought not to exist (deepwater, Brazil's pre-salt) added recoverable oil, lifting and stretching the curve.
Better extraction technology	Horizontal drilling and hydraulic fracturing unlocked US shale / tight oil from ~2010, transforming the supply picture and delaying the supposed peak.
Unconventional sources	Oil sands, tight oil, coal seam gas and oil shale count as oil-equivalent supply once technology and prices allow.
Price	High prices make difficult, low-grade fields economic, so reserves that were "uneconomic" suddenly count.
Demand	Efficiency, EVs and renewables can lower demand — which, as we'll see, can bring the peak forward from the other side.

✓ TIP

The deeper cost: EROEI Even when technology delays the peak, it rarely makes oil cheaper in energy terms. EROEI (energy return on energy invested) falls as we chase harder oil — you must burn more energy to win each barrel. A gusher of easy crude might return 100:1; tar sands might return closer to 5:1. Falling EROEI is a quieter, more reliable signal than any single "peak year."

B3 Peak supply vs peak demand

The original peak-oil idea was a supply story: production peaks because the resource is finite. But the conversation has flipped. Many analysts now argue the peak may arrive because demand falls away first — driven by electric vehicles, efficiency, renewables and climate policy — rather than because we can't pump any more.

◆ ANALOGY

Analogy — why did the party end? Did the music stop because the band ran out of songs (peak supply) or because the crowd went home (peak demand)? Both empty the dancefloor — but for opposite reasons. Modern peak-oil debate is mostly about which one happens first.

As of 2026 the timing is genuinely contested: some major outlooks see oil demand reaching a plateau later this decade as EV sales and efficiency bite, while others project demand still growing for years yet. For the exam, the safe, sophisticated line is: peak oil is no longer just about supply running short — it is increasingly about demand falling away, and credible forecasts disagree on the timing. Showing you know the debate exists is worth more than betting on one date.

◆ NOTE

Why this matters for energy choices If you believe oil will peak on the supply side, you prepare for scarcity and price shocks. If you believe it peaks on the demand side, you prepare for stranded assets — expensive oil infrastructure built just as the world stops needing it. Either way, the policy conclusion points the same direction: build the transition early.

★ Bass Strait: a Hubbert curve on Victoria's doorstep

You don't need to look overseas for peak oil. Off the Gippsland coast — the view from our case-study town, Tarra Cove — the real Bass Strait fields tell the whole story in miniature.

The Gippsland Basin (Bass Strait), run by Esso/ExxonMobil with Woodside, has produced oil and gas for more than 50 years. Its oil output rose, peaked and is now in mature decline — about half of all the wells no longer produce, and decommissioning of the spent platforms begins in 2027. A textbook Hubbert curve, right here.

As local crude declined, Australia came to import roughly 80–90% of its liquid fuel. Two refineries closed in 2021 (Kwinana in WA and Altona in Victoria), leaving just two nationally — Viva Energy in Geelong and Ampol's Lytton in Brisbane. That raises real fuel-security questions when global supply is disrupted.

✓ TIP

Use this in the exam If a question asks for a real example of peak oil or of changing fossil-fuel use, Bass Strait / the Gippsland Basin is a high-value Australian case: a maturing field in clear decline, driving a national shift from local production to imported fuel, with platform rehabilitation now underway (which links forward to KK08).

02 Decode the command word

◆ COMMAND WORDS

The verb tells you what shape your answer must take. Tap one to see what KK04 expects.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

03 Six exam traps

⚠ EXAM TRAPS

Every one of these has cost real students real marks. Read them as a pre-flight checklist.

⚠ WATCH OUT

Trap 1 — "Peak oil means we're running out" The single biggest error. Peak oil is the peak of the extraction rate; about half the resource remains at the peak. Say rate, not "amount left".

⚠ WATCH OUT

Trap 2 — Describing the curve instead of explaining it "Explain" needs causes. Don't just say "use went up then levelled off" — name drivers (industrialisation, transport, then efficiency, renewables, policy).

⚠ WATCH OUT

Trap 3 — Treating "fossil fuels" as one thing Coal, oil and gas have different time-trajectories. Coal led and is plateauing; gas is still growing. Distinguish them where the marks allow.

⚠ WATCH OUT

Trap 4 — Forgetting why predictions kept changing If asked why the global peak shifted, you must mention discoveries, technology (fracking/shale), unconventional sources and price — not just "they were wrong".

⚠ WATCH OUT

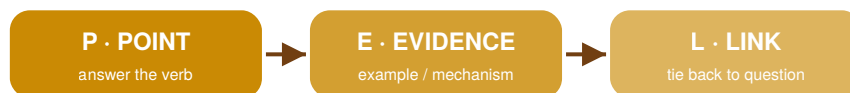
Trap 5 — Missing the supply-vs-demand flip Top answers note that peak oil may now be driven by falling demand (EVs, efficiency, policy), not only by limited supply.

⚠ WATCH OUT

Trap 6 — No real example A named case (Bass Strait decline, US 1970 peak, or the US shale surge) lifts a generic answer into a high band. Keep one ready.

MARKS

04 P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

📝 P·E·L WRITING

Build a high-band answer in three moves — Point, Evidence, Link. Type into each box; the trainer flags the key ideas examiners look for.

05 Same question, two answers

Q: "Define peak oil and explain why it does not mean the world is about to run out of oil." (3 marks)

"Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use."

"Peak oil is the point at which the maximum rate of oil extraction is reached, after which production permanently declines. It does not mean oil is running out, because at the peak roughly half the recoverable resource still remains underground. What declines is the rate of production, as the cheap, easy oil is used first and the remainder is harder and costlier to extract."

06 Practice exam questions

◆ NOTE

Five VCAA-style questions · 20 marks total Write your answer, reveal the marking guide, then self-mark. Your score saves automatically and feeds the dock.

- Improved extraction technology — horizontal drilling and hydraulic fracturing unlocked shale/tight oil that earlier models assumed was unrecoverable, adding supply.

- New discoveries — fields such as deepwater and pre-salt reserves added recoverable oil not counted before.
- Unconventional sources — oil sands and tight oil became viable, expanding the resource base.
- Higher prices — made previously uneconomic, harder-to-reach fields worth extracting.
- Define / frame: peak oil = peak of the extraction rate; traditionally a supply -side idea (finite resource → Hubbert curve).
- Supply still matters: oil is finite, EROEI falls as easy oil is used up, and supply shocks still drive price spikes — so a supply-led peak is not impossible.
- Demand-side case: EVs, efficiency, renewables and climate policy are slowing the rate of oil use; several major outlooks now see demand plateauing later this decade, suggesting a demand-led peak.
- Uncertainty: credible forecasts disagree on timing — some still project growth — so any single date is contestable.
- Judgement: a defensible conclusion (e.g. that the balance has shifted toward demand in wealthy economies, while supply constraints still bite globally), explicitly supported by the points above.

Fossil fuel use & peak oil

"Changes in the rate of fossil fuel use over time and the concept of peak oil."

Overview

This dot point has two halves, and the exam rewards you for handling both cleanly. First, how the rate of fossil fuel use has changed over time — the long arc from coal to oil to gas, and why that arc is now bending. Second, the concept of peak oil — a specific idea about the rate of extraction from a finite resource, which is almost always tested through one killer misconception.

✓ TIP

The one-line version Use over time: fossil fuel use grew explosively across the 20th century and is now plateauing in places. Peak oil: the moment the rate of oil extraction tops out and begins a permanent decline — not the moment oil runs out.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use." Why it loses marks: the definition is wrong — it equates peak oil with running out, the exact trap. No mention of extraction rate, no "half remains" idea, and the second sentence is vague geography rather than the real reason.

✓ STRONG / MODEL ANSWER

Peak oil is the point at which the maximum rate of oil extraction is reached, after which production permanently declines. It does not mean oil is running out, because at the peak roughly half the recoverable resource still remains underground. What declines is the rate of production, as the cheap, easy oil is used first and the remainder is harder and costlier to extract." Why it scores: precise definition built on rate; directly tackles the misconception; uses the half-remains fact; explains the mechanism (easy oil first). Three clean, separable marking points.

Where does the energy actually go?

THE BIG IDEA

Energy is never used up — it's transformed, and degraded

A power station doesn't make energy. It converts energy that's already stored in something (chemical bonds in coal, water held high behind a dam, moving air) into the form we want: electrical energy.

The First Law of Thermodynamics says energy can't be created or destroyed — only changed from one form to another. So if you put 100 units of energy in, 100 units come out. The catch is which forms they come out as. We want electricity. What we usually also get is a pile of low-grade heat spilling into the air and water — energy that's still real, but useless for running a fridge.

An exam conversion chain is built from these forms. Learn the colours — we use them all through this module.

Energy stored in bonds — coal, gas, biofuel, food, batteries. Released by combustion or reaction.

Stored by position — water held high behind a dam. Gravitational PE = mgh .

Energy of motion — falling water, moving air, spinning turbine blades. KE = $\frac{1}{2}mv^2$.

The total working energy of moving machinery (kinetic + potential of the moving parts) — e.g. a turbine shaft.

Heat — random motion of particles. The form most energy wastes into. Hot steam, hot gases.

The useful output we're chasing — energy carried by moving charge through the grid.

◆ ANALOGY

Analogy — the leaky-bucket relay Picture a fire brigade passing buckets of water hand to hand. Every single handover spills a little. With one handover you lose a splash; with four handovers you lose four splashes. The water that reaches the fire is the useful output; the puddles on the ground are the wasted heat. The more conversions in the chain, the more total spillage — that's exactly why multi-step generation is less efficient.

THE ONE FORMULA

Efficiency = useful out ÷ total in

This is the single equation behind every calculation in this dot point. Memorise it, and memorise that the answer can never exceed 100%.

• NOTE

η ("eta") = efficiency. Works with energy (J, kJ, MJ, GJ) or power (W, kW, MW) — as long as both top and bottom use the same units.

A question might give you energy (joules — a total amount) or power (watts — a rate, joules per second). The formula is identical. Just keep the units consistent.

By the First Law, everything in must come out. So the energy that isn't useful is the rest — almost always lost as thermal energy (heat) to the surroundings.

The generator at Yarrabin's hydro plant receives 500 MJ of kinetic energy from the spinning turbine and delivers 475 MJ of electrical energy. What is its efficiency?

Wasted: $500 - 475 = 25$ MJ lost as heat in the windings and bearings. Notice the answer is below 100% — it always is.

INTERACTIVE · SCENE 02

The conversion-chain lab

Each box shows the energy surviving after that conversion. The overall efficiency is the running product — multiply the step efficiencies, never add them.

THE EXAM-CRITICAL SKILL

Multi-step: why the losses compound

When energy passes through several conversions, you multiply the step efficiencies (as decimals) — you never add or average them.

• NOTE

Then $\times 100$ to express as a percentage. The overall figure is always lower than the worst single step.

◆ ANALOGY

Analogy — three middlemen each take a cut Say your pocket money passes through three people, and each keeps a slice. The first lets 90% through, the second 50%, the third 95%. You don't subtract their cuts — you keep $0.90 \times 0.50 \times 0.95 = 0.4275$, i.e. about 43%. More middlemen (more conversion steps) means less reaches you, fast.

Coal makes electricity through a four-form chain. Each conversion has its own efficiency:

So roughly two-thirds of the coal's energy is thrown away — mostly as waste heat at the steam-turbine step. Real Victorian brown coal is even lower (~25–30%) because wet brown coal carries a lot of water that must be boiled off first.

The killer step above is thermal $\rightarrow$ kinetic (only 42%). That's the heat-engine bottleneck — turning heat into motion is fundamentally limited by the Second Law of Thermodynamics (the next dot point, KK06). Any source that burns something — coal, gas, biomass, even nuclear — is stuck with this step.

Hydro and wind don't burn anything. Their energy is already mechanical, so they skip the thermal stage entirely:

Source	Conversion chain	Typical overall efficiency
Hydro	Potential $\rightarrow$ Kinetic $\rightarrow$ Mech $\rightarrow$ Elec	~90%
Wind	Kinetic $\rightarrow$ Mech $\rightarrow$ Elec	~40%*
Gas (CCGT)	Chem $\rightarrow$ Thermal $\rightarrow$ Kinetic $\rightarrow$ Elec	~55%
Black coal	Chem $\rightarrow$ Thermal $\rightarrow$ Kinetic $\rightarrow$ Elec	~38%
Brown coal	Chem $\rightarrow$ Thermal $\rightarrow$ Kinetic $\rightarrow$ Elec	~28%
Solar PV	Light $\rightarrow$ Elec (1 step)	~20%

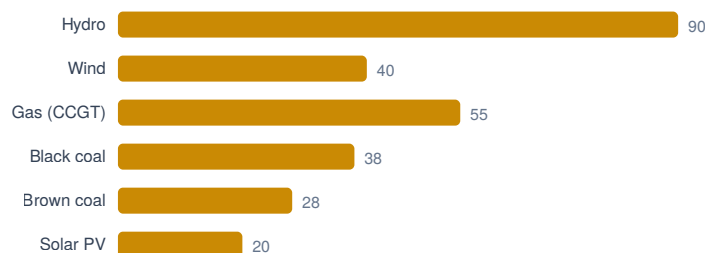


Figure · Source compared at a glance.

*Wind is one step short but limited by the Betz limit — a turbine can extract at most 59.3% of the wind's kinetic energy. Solar PV has just one conversion yet sits low because of the physics of the photovoltaic cell, not because of a long chain. Lesson: fewer steps usually helps, but each step's own ceiling matters too.

INTERACTIVE · SCENE 03

Build-your-own Sankey diagram

A Sankey diagram draws energy as a flow whose width is proportional to the amount. The useful stream stays on the main path; wasted heat branches off. Drag the three step efficiencies and watch the river narrow.

Try dragging step 2 (the turbine) up to 100% — see how much the whole river fattens. That single step is the heat-engine bottleneck doing all the damage. Examiners love asking you to read useful vs wasted straight off a Sankey.

WORKED EXAMPLES

See the marks being earned

Three classic question types. Open each, cover the working, and try it first.

Yarrabin's gas plant converts chemical energy → thermal (88%) → kinetic (60%) → electrical (96%). Calculate the overall efficiency. (2 marks)

Marking: 1 mark for multiplying (not adding) the three decimals; 1 mark for the correct answer with a % sign. Round sensibly and keep the unit.

The solar farm receives 1500 MJ of light energy in an hour and operates at 22% efficiency. Determine the electrical energy generated and the energy wasted. (3 marks)

Marking: rearranging the formula correctly (1), the 330 MJ output (1), the 1170 MJ waste by subtraction (1). Note the waste is not destroyed — First Law — it leaves as heat.

A biomass plant has an overall efficiency of 30%. Its boiler (chem→thermal) is 85% efficient and its generator (kinetic→elec) is 94% efficient. Show that the turbine step is about 38% efficient. (3 marks)

Marking: set up the product (1), isolate the unknown by dividing (1), correct value matching the "show that" target (1). For a "show that", you must end at the stated number.

DECODE THE QUESTION

What the command word is really asking

◆ COMMAND WORDS

In calculation questions, the verb tells you exactly how much working to show. Match your response to it. Reach a numerical answer with units. Write each line of working — marks live in the method, not just the final number. Same as calculate — find the value, showing the steps that get you there. You're given the answer. Your job is the working that lands on it. Don't skip steps; finish on the stated value. State a figure for each source and an explicit point of difference (e.g. "hydro ~90% vs coal ~30% because hydro skips the thermal step"). Give the reason or mechanism — usually "energy is lost as heat at each conversion, and losses multiply." Link cause to effect. Name the form or value. No working needed — but be precise (say "thermal energy", not "energy").

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

DON'T LOSE EASY MARKS

Six traps that cost real marks

⚠ EXAM TRAPS

0.36 is a fraction; 36% is a percentage. Decide once whether you're working in decimals or percent and stay there. Fix: work in decimals, convert to % only at the very end. Putting total input on top gives an efficiency above 100% — impossible. Fix: useful output always goes on top; the answer must be $\leq 100\%$. The ratio only works if top and bottom share units. Fix: convert first — 1 kWh = 3.6 MJ — then divide. It contradicts the First Law and examiners penalise it. Fix: say energy is transferred to the surroundings as heat or degraded to a less useful form. Efficiency = how much of the input becomes electricity. Capacity factor = how much of the time a plant runs near full output. A wind farm can be ~40% efficient and have a ~35% capacity factor — different ideas. Fix: read which one the question wants.

WRITING TRAINER

The P·E·L move for "explain" parts



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P-E-L WRITING

Calculation questions often end with a worded part — "explain why multi-step generation is less efficient." Examiners want a Point, Evidence, and a Link. Type an answer and watch the checklist light up. "A coal plant converts energy through four forms — chemical → thermal → kinetic → electrical — and energy is transferred to the surroundings as heat at each conversion. Because the step efficiencies multiply, the overall efficiency (~30%) is much lower than hydro. Hydro skips the chemical-to-thermal heat-engine step entirely, going potential → kinetic → mechanical → electrical, so it avoids the largest loss and reaches ~90%. Fewer steps and no heat engine mean hydro keeps far more of its input as useful electricity." Why it scores: names the forms, identifies the heat-engine step as the bottleneck, uses the "efficiencies multiply" rule, gives comparative figures, and links explicitly back to the comparison asked.

PRACTICE EXAM · 20 MARKS

Five VCAA-style questions

★ PRACTICE & SELF-MARK

Work each one on paper first. Then reveal the weak answer, the strong answer and the marking guide, and award yourself a score — it saves to your radial dock.

- 1 mark — correct ratio $320 \div 800$ (useful over total).
- 1 mark — answer of 40% with the % sign.
- 1 mark — multiplies the three decimals (method).
- 1 mark — overall efficiency $\approx 32\%$.
- 1 mark — ≈ 646 MJ electricity (0.323×2000).

★ PRACTICE & SELF-MARK

"The student is wrong because the plant is only 30% efficient, so 70% is lost. That's just how power stations work." The energy is not destroyed. The First Law of Thermodynamics states energy cannot be created or destroyed, only transformed. In the coal plant, chemical energy becomes thermal, then kinetic, then electrical. The "wasted" 70% is real energy that has been transferred to the surroundings as low-grade thermal energy (heat) — in the cooling water, flue gases and friction — at each conversion step. It is degraded to a less useful form, not gone.

- 1 mark — energy is not destroyed / conserved.
- 1 mark — cites the First Law of Thermodynamics.
- 1 mark — names forms (chemical→thermal→kinetic→electrical).
- 1 mark — the waste is thermal energy transferred to the surroundings.

★ PRACTICE & SELF-MARK

"Hydro is better because it's renewable and cleaner. Coal is bad for the environment and only 30% efficient. Hydro is 90%." Hydro chain: gravitational potential → kinetic (falling water) → mechanical (turbine) → electrical. There is no thermal step, so it avoids the big heat-engine loss and reaches ~90%. Coal chain: chemical → thermal → kinetic → electrical. The chemical-to-thermal-to-kinetic stage is a heat engine, which loses a large fraction as waste heat (limited by the Second Law), pulling the overall to ~30%. Because step efficiencies multiply, coal's extra thermal step compounds the losses, so hydro keeps roughly three times as much of its input as useful electricity.

- 1 mark — correct hydro chain with named forms.
- 1 mark — correct coal chain with named forms.
- 1 mark — identifies the thermal / heat-engine step as the key loss.
- 1 mark — links to losses multiplying across steps.
- 1 mark — explicit comparative conclusion tied to the figures.

★ PRACTICE & SELF-MARK

"(a) Gas is about 90% because that's the biggest number. (b) Gas needs less energy. (c) Gas is better because it's more efficient." (c) By efficiency, gas wins — it needs far less input per unit of electricity. But efficiency alone is misleading: the solar input is free, renewable sunlight, whereas the gas input is a finite fossil fuel that releases CO₂. A low-efficiency renewable can still be the more sustainable choice.

- 1 mark — (a) multiplies decimals → ≈ 54%.
- 1 mark — (b) gas input ≈ 75 MJ (40 ÷ 0.536).
- 1 mark — (b) solar input ≈ 190 MJ (40 ÷ 0.21).
- 1 mark — (c) states which is more efficient with reason.
- 1 mark — (c) names a limitation of efficiency alone (e.g. free/renewable input, emissions).
- 1 mark — overall coherence: correct rearrangement of $\eta = \text{useful} \div \text{input}$ throughout.

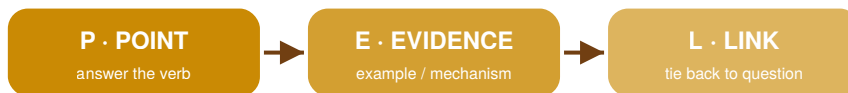
✦ KEY TAKEAWAY

Five things to walk into the exam knowing $\eta = \text{useful output} \div \text{total input} \times 100$. Useful goes on top; the answer is always $\leq 100\%$. Multi-step efficiencies multiply (as decimals) — never add or average them. Wasted energy = input – useful, and it's transferred to the surroundings as thermal energy (heat) — never "lost" or "destroyed" (First Law). The thermal step is the bottleneck. Anything that burns fuel runs a heat engine, limited by the Second Law — that's why coal/gas sit low and hydro/wind sit high. Efficiency isn't everything — a low-efficiency renewable with a free, clean input can still be the more sustainable choice.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

800 – 320 = 480, so it's 480 kJ efficient." Confuses the wasted energy with efficiency, gives no ratio, no percentage. This is a wasted-energy calculation, not an efficiency one.

✓ STRONG / MODEL ANSWER

$\eta = \text{useful output} \div \text{total input} \times 100 = 320 \div 800 \times 100 = 40\%$. Correct formula, substitution shown, units cancel, percentage stated.

✗ WEAK ANSWER

85 + 40 + 95 = 220, divided by 3 = 73% efficient. So $0.73 \times 2000 = 1460$ MJ." Averaging the percentages is the classic multi-step error — efficiency can't be found by adding or averaging. Both the method and both numbers are wrong.

✓ STRONG / MODEL ANSWER

$\eta_{\text{overall}} = 0.85 \times 0.40 \times 0.95 = 0.323 = 32\%$ ($\approx 32.3\%$). Electricity out = $0.323 \times 2000 = \approx 646$ MJ. Multiplies decimals, converts to %, then applies it to the input.

✗ WEAK ANSWER

The student is wrong because the plant is only 30% efficient, so 70% is lost. That's just how power stations work." Repeats the misconception ("lost"), names no forms, cites no law, and doesn't correct the "destroyed" claim. Only the efficiency figure earns anything.

✓ STRONG / MODEL ANSWER

The energy is not destroyed. The First Law of Thermodynamics states energy cannot be created or destroyed, only transformed. In the coal plant, chemical energy becomes thermal, then kinetic, then electrical. The "wasted" 70% is real energy that has been transferred to the surroundings as low-grade thermal energy (heat) — in the cooling water, flue gases and friction — at each conversion step. It is degraded to a less useful form, not gone. Corrects the error, names forms, cites the First Law, explains where the energy actually goes.

The Two Laws That Decide Every Energy Choice

WHY A PHYSICS RULE BELONGS IN ENVIRONMENTAL SCIENCE

01 Why a physics rule belongs in Environmental Science

The study design asks you to explain the implications of the first and second laws of thermodynamics when making energy choices. Not the maths of efficiency (that was KK05) — the consequences. These two laws are the reason a coal plant throws away two-thirds of its fuel, the reason "free energy" machines are scams, and the reason cutting waste beats building more supply.

◆ NOTE

The Tarra Cove decision. The town of Tarra Cove, on the edge of Bass Strait, has to power its new desalination plant. The council is handed three options: pipe in electricity from the inland Wirndook brown-coal station, build local solar + battery, or run a gas peaker. Every councillor wants the "most efficient" choice — but none of them can escape the two laws. By the end of this module, you'll be the one explaining why.

THE TWO LAWS, STATED CLEANLY

02 The two laws, stated cleanly

Learn both the formal statement (for the high-mark questions) and the plain-English version (so you actually understand it).

"Energy cannot be created or destroyed, only transformed from one form into another. The total energy of an isolated system is constant."

Whatever energy you start with, you still have — it has just changed form. If only 33 units come out useful, the other 67 didn't vanish; they became low-grade heat spread into the surroundings.

"In any energy conversion, the total entropy (disorder) of the system and surroundings increases. Some energy is always degraded to low-grade, dispersed heat that can no longer do useful work."

You can account for all the energy (first law), but you can never use all of it. A bit becomes useless waste heat every single step, and it only flows one way: hot → cold, ordered → disordered.

◆ NOTE

The one-line memory hook: First law = you can't win (you'll never get out more than you put in). Second law = you can't break even (you'll always get out less than you put in). Together: you can't get out of the game — there's no perfect machine.

FIRST LAW: THE ENERGY IS STILL THERE CONSERVATION

03 First law: the energy is still there

The most common misread is thinking inefficiency means energy "lost". It isn't lost — it's relocated and reshaped. When the Wirndook turbine is 40% efficient, the missing 60% is now warm cooling water in the Kalduke River, warm flue gas up the stack, and friction heat in the bearings. Add it all up and you're back to 100%.

◆ ANALOGY

Analogy · the energy bank account Think of energy like money in a sealed economy. You can move it between accounts (chemical → heat → motion → electricity) but you can never print new notes or burn them out of existence. Every cent is always somewhere. An "energy audit" of any machine must balance to exactly 100% — if it doesn't, you've missed a stream of waste heat.

The implication for choices: there is no such thing as a free lunch. Any device claiming to output more energy than it consumes — an "over-unity" or "perpetual motion" generator — breaks the first law and cannot exist. When an energy pitch sounds too good, the first law is your fraud detector.

◆ NOTE

Where does the energy "go" in your phone charger? A charger pulling 10 W but delivering 8 W to the battery isn't destroying 2 W — that's why the plug feels warm. The first law says those 2 W must appear somewhere, and the second law says "somewhere" is almost always low-grade heat.

SECOND LAW: ENERGY QUALITY ONLY EVER FALLS ENTROPY ↑

04 Second law: energy quality only ever falls

The first law treats every joule as equal. The second law says they're not. High-grade energy is concentrated and ordered — it can do work (electricity, the chemical energy in petrol, a tank of compressed steam). Low-grade energy is dispersed and lukewarm — heat spread thinly through the air or ocean. Every conversion shifts a slice of high-grade into low-grade, and the universe's total entropy (disorder) rises.

Crucially, this is a one-way street. Heat flows from hot to cold on its own, but never the reverse without you spending more high-grade energy (that's what a fridge or heat pump does — it pumps heat uphill, but only by consuming electricity). You cannot un-disperse waste heat back into a neat tank of useful energy for free.

Start: energy is tightly packed and hot in one corner — it can do work. Press Let it run to watch the second law happen.

◆ ANALOGY

Analogy · the drop of cordial Drip red cordial into a glass of water and it spreads until the whole glass is pale pink. It never spontaneously gathers back into a single drop. Energy does the same: concentrated → spread out, and never spontaneously the reverse. Stirring it back together would cost you energy — and you'd still never get the clean drop returned. That irreversibility is the second law.

WHY NO POWER STATION IS EVER 100% EFFICIENT

05 Why no power station is ever 100% efficient

Put the two laws together and you get the headline result of this whole chapter. Any source that works by burning fuel to make heat, then converting heat to motion — coal, gas, oil, nuclear, even biomass — is a heat engine, and heat engines are hard-capped by the second law. You can only extract work from heat as it flows from a hot source to a cold sink, and a large fraction must be dumped to the cold side as waste. That waste isn't bad engineering; it's physics.

Energy pathway	Heat engine?	Typical real efficiency	Why
Brown-coal electricity (Wirndook)	Yes	~25–33%	Wet, low-energy lignite + unavoidable heat-engine losses
Combined-cycle gas	Yes	~50–60%	Captures more of the heat in two stages, but still capped
Solar PV	No	~15–22%	Not a heat engine, but photons missed/lost as heat
Hydro / wind	No	~80–90%	Converts existing motion directly — very few losses
Solar hot water (direct)	No	~60–70%	Sunlight → heat → hot water, almost no conversion steps

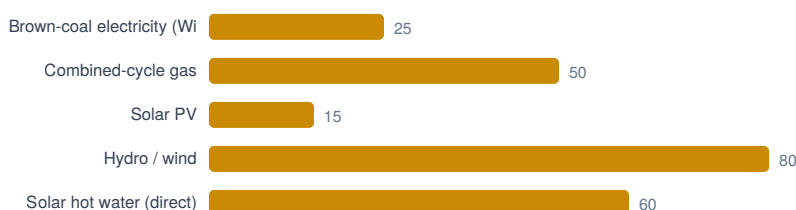


Figure · Energy pathway compared at a glance.

◆ NOTE

Read the table like an examiner. Notice the heat engines cluster low and the direct converters cluster high. That isn't a coincidence — it's the second law sorting them. A strong exam answer names both laws: the first explains where the missing energy went (waste heat), the second explains why it's unavoidable (entropy must rise).

ENERGY QUALITY: MATCH THE TOOL TO THE JOB

06 Energy quality: match the tool to the job

Because the second law makes high-grade energy precious, a smart energy choice matches the quality of the energy to the quality the job actually needs. Using premium electricity to do a job that only needs low-grade warmth is like using filtered drinking water to flush a toilet — it works, but you've wasted something valuable.

Electricity, chemical fuels, compressed gas. Can be converted into almost any form. The most useful and most "expensive" in second-law terms — every conversion that made it already paid a tax.

Waste heat from a power station, lukewarm air, the warmth in the ocean. Plenty of it exists, but it's spread too thin to drive a turbine or a motor. Mostly destined to be lost.

The clever move is cogeneration (also called combined heat and power): instead of fighting the waste heat from a power station, you capture it and pipe it to heat nearby buildings or run an industrial process. You can't beat the second law — but you can stop throwing the loser's share away.

THE IMPLICATIONS — MAKING THE TARRA COVE CHOICE

07 The implications — making the Tarra Cove choice

This is the exact skill the study design names: turning the two laws into decisions. Every conversion step in a chain pays a second-law tax, and the taxes multiply. So one rule dominates: fewer conversions, and use the source closest to the form you need.

The five practical implications you should be able to argue in an exam:

- Each step sheds waste heat (2nd law), so chains with fewer links keep more useful energy. Direct solar hot water beats coal-fired resistive heating for the same warmth.
- Coal, gas, oil and nuclear are capped low by the second law. If a choice is dominated by heat engines, large unavoidable losses are locked in.
- Don't burn high-grade electricity for low-grade warmth. A heat pump moving existing heat beats a resistive heater making new heat.
- You can't stop the loss, but you can reuse it for heating instead of dumping it to a river or the sky.
- Every joule not used avoids all the upstream conversion losses — so efficiency and reduced demand do more than new supply.
- "Over-unity" and perpetual-motion claims break the first law — no investigation needed, the physics already says no.

WHICH LAW IS IT? — SORT THE STATEMENTS

08 Which law is it? — sort the statements

◇ ACTIVE RECALL

The single most common exam slip is naming the wrong law. Tap a statement, then tap the bin it belongs in. First = about energy being conserved / where it went. Second = about quality, waste heat, entropy or direction. Both = needs the two laws together.

DECODE THE COMMAND WORD

09 Decode the command word

◆ COMMAND WORDS

The verb in the question tells you how much to write. Tap each to see what the marker wants for a thermodynamics question.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

SIX TRAPS THAT COST REAL MARKS

10 Six traps that cost real marks

⚠ EXAM TRAPS

Each of these has burned students in real VCAA-style answers. Open them and steal the fix.

P·E·L WRITING TRAINER

11 P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔑 P·E·L WRITING

P oint → E xplain → L ink. Write a 3-mark answer below. As you type, the targets light up when you hit the ideas a marker is hunting for. This is live keyword detection — it rewards substance, not length.

SAME QUESTION, TWO ANSWERS

12 Same question, two answers

A weak answer and a strong answer to the same 4-mark prompt. The highlights show exactly where marks are won and lost.

Question. The Tarra Cove council is choosing between coal-fired electricity (then used in resistive heaters) and a direct solar hot-water system to warm water for its swimming pool. Using the laws of thermodynamics, justify which is the more efficient choice. (4 marks)

• NOTE

Names no law. "Renewable / doesn't pollute" is a different dotpoint — it doesn't answer a thermodynamics question. "Wastes energy" is asserted, never explained. No mention of conversion steps or energy quality.

• NOTE

Names both laws and uses them correctly, tracks the conversion chain, invokes energy quality matching, and lands a clear justified decision. Full marks.

PRACTICE EXAM QUESTIONS / 20 MARKS

13 Practice exam questions

★ PRACTICE & SELF-MARK

Five VCAA-style questions, scaling from 2 to 6 marks. Write your answer first, reveal the model answer and marking guide, then award yourself stars. Your scores save automatically and feed the dock below.

14 One-screen cheat sheet

- Energy is conserved — created/destroyed by nothing, only transformed.
- Inefficiency $\neq$ loss; the energy became low-grade waste heat in the surroundings.
- Implication: no perpetual motion / over-unity machine can exist.
- Every conversion increases entropy; some energy degrades to dispersed heat.
- Energy quality always falls; heat flows hot $\rightarrow$ cold, never the reverse for free.
- Implication: $\eta < 100\%$ always; heat engines are capped low.
- Fewer conversion steps = more useful energy kept.
- Match quality to task — don't waste electricity on warmth.
- Cogeneration reuses waste heat; saving energy skips all upstream losses.
- First = where did it go; Second = why can't I reuse it.
- "Renewable / clean" is a different dotpoint — don't answer with it here.
- Always name the law before you use it.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Solar is better because it's renewable and doesn't pollute. Coal is bad for the environment and wastes a lot of energy. The solar option is more efficient and cheaper so the council should pick it. Names no law. "Renewable / doesn't pollute" is a different dotpoint — it doesn't answer a thermodynamics question. "Wastes energy" is asserted, never explained. No mention of conversion steps or energy quality.

✓ STRONG / MODEL ANSWER

Heating water needs only low-grade thermal energy. The coal path makes electricity through many conversions (chemical $\rightarrow$ heat $\rightarrow$ motion $\rightarrow$ electricity $\rightarrow$ heat), and by the second law each step degrades some energy to waste heat, so only $\sim 33\%$ of the coal's energy survives. The first law confirms the rest isn't destroyed — it's lost as heat to the surroundings. The solar system converts sunlight directly into heat in far fewer steps, so much less energy is degraded. Because it matches a low-grade need with a low-grade source, the solar option is the more efficient choice. Names both laws and uses them correctly, tracks the conversion chain, invokes energy quality matching,

Sustainability principles, applied across the energy supply chain

Every other dot point in this chapter has handed you a piece of an energy source — what it is (KK01–02), what burning it does to the carbon cycle (KK03), how fast we are using it (KK04), how efficiently it converts (KK05), and what the laws of thermodynamics permit (KK06). KK07 is where you assemble those pieces into a judgement. It asks you to run the six sustainability principles like a checklist across the five stages an energy resource passes through on its way from the ground to your power point.

THE SUPPLY CHAIN

01 Five stages, one resource

The study design names the stages precisely: accessing → extracting → processing → transporting → using. Learn them as a sequence, because exam questions love to pin a principle to one specific stage ("evaluate the user-pays principle at the extraction stage").

The scene below follows a single energy resource down the Tarra Cove corridor in Gippsland — the fictional coastal town we have used across this chapter, sitting beside the real Latrobe Valley brown-coal fields and the Bass Strait gas province. Click a stage to fly the camera to it and read what each stage physically does before we judge it.

• NOTE

3D view unavailable in this browser — the stage cards and readout below work exactly the same.

THE YARDSTICK

02 The six sustainability principles

These are the same six VCAA principles you met in U3-O2 — but here you swap "environmental management" for "energy". Memorise the one-line "energy test" on each card: that is the lens you point at every stage.

Two principles students confuse every year. Intergenerational equity is fairness between generations — today's energy use must not leave a degraded planet or an un-rehabilitated mine for people not yet born.

Intragenerational equity is fairness within today's generation — energy access, costs and harms shared fairly between rich and poor, city and region, here and overseas. Same root word (equity = fairness); different time direction. Inter = between (generations); intra = within (one generation).

THE WHOLE PICTURE

03 Principle × stage: the interactive matrix

This grid is the dot point. Six principles down the side, five stages across the top. Click any cell to see how that principle applies at that stage, with an Australian example. This is the structure to reach for whenever a question says "apply the sustainability principles to...".

REAL AUSTRALIAN CASES

04 Examples that earn marks

Examiners reward specific, current, Australian examples over vague generalities. Lock these three — one each for the principles students find hardest to illustrate.

Victoria permanently banned fracking and coal-seam-gas exploration and wrote the ban into the state constitution (2021). Even though the science on groundwater risk was contested, the government acted to prevent serious, potentially irreversible harm to farmland and aquifers. That is the precautionary principle at the accessing/extracting stage.

Australia's Safeguard Mechanism caps the emissions of facilities over 100,000 t CO₂-e/yr; baselines fall ~4.9% a year and emitters over the line must buy carbon credits (~\$37/t in 2026). Polluters pay for the climate cost of processing & using fossil energy — instead of dumping it on the public.

The Hazelwood mine (closed 2017) is being rehabilitated into a full pit lake — a "safe, stable, sustainable, non-polluting" landform — so a 135 m-deep void with fire and collapse risk is not inherited by future generations. Rehabilitation bonds and a 2026 trailing-liabilities law force the company, not future taxpayers, to pay.

The user-pays / intergenerational tension to remember: Engie's Hazelwood rehabilitation bond is about \$289 million, but the estimated full cost is roughly \$743 million. When the bond is smaller than the real bill, the shortfall risks falling on the public and the future — a live example of a principle being only partly upheld. "Partly upheld, because..." is exactly the nuance that lifts an evaluate answer.

COMPARE TWO PATHWAYS

05 Brown coal vs solar + storage

"Compare" and "evaluate" questions want you to weigh one source against another across the principles. Toggle the two pathways and watch the pressure each one puts on each principle. Taller bar = more strain on that principle (worse for sustainability).

• NOTE

3D view unavailable — the principle bars below update the same way.

ACTIVE RECALL

06 Which principle is being upheld?



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

READ THE QUESTION

07 Command-word decoder

◆ COMMAND WORDS

The verb tells you what shape your answer must take. Mis-reading "evaluate" as "describe" is the single most common way to lose marks on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

DON'T FALL FOR THESE

08 Six exam traps

⚠ EXAM TRAPS

Each of these has cost real students real marks. Open one, then close it and try to recall the fix.

P · E · L

Build a body paragraph that scores

High-band answers on this dot point follow P oint → E vidence → L ink. Make a point (name the principle & stage), give evidence (a specific Australian example with a figure), then link back to "upholding" or "undermining" the principle. Type below — the meter lights up as you hit each move.

± WORKED EXAMPLE

Same question, two answers

Q. "Apply one sustainability principle to the extraction stage of an energy resource of your choice." (3 marks) — watch how the weak answer describes, while the strong answer applies and links.

Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation.

• NOTE

Why it loses marks: names principles but never applies them to the extraction stage; no Australian example; "bad for the environment" is not ecological integrity; lists three principles vaguely instead of developing one.
Mark: identifies a principle (1), but no application and no evidence.

The user-pays principle requires those who extract a resource to bear its full environmental cost. At the extraction stage of Victorian brown coal, the operator of a Latrobe Valley mine must lodge a rehabilitation bond with the State so that the cost of returning the 135 m-deep void to a safe, stable landform is paid by the miner, not the public. This upholds user-pays by internalising the rehabilitation cost into the price of the coal — though it is only partly upheld where the bond (\$289M) is below the estimated full cost (\$743M).

• **NOTE**

Why it scores: defines the principle (1), applies it to the named extraction stage with a specific Australian mechanism + figure (1), and links explicitly to "uphold", with a nuance that shows evaluation (1).

★ **PRACTICE EXAM**

Five VCAA-style questions · 20 marks

★ **PRACTICE & SELF-MARK**

Attempt each in your logbook first, then reveal the model answer and marking scheme. Award yourself marks honestly — the dock at the bottom tallies your total and saves it on this device.

≡ **ONE-SCREEN SUMMARY**

KK07 cheat sheet

- Accessing — finding it & getting legal/physical entry
- Extracting — mining, drilling, or harvesting the flux
- Processing — refining, drying, enriching, converting
- Transporting — pipelines, transmission, ships, trucks
- Using — final conversion to heat / motion / electricity
- Ecological integrity — protect ecosystems & their function
- Resource-use efficiency — most useful energy per unit input
- Intergenerational equity — fair to future generations
- Intragenerational equity — fair within today's generation
- Precautionary — act on serious/irreversible risk under uncertainty
- User-pays — those who use/pollute bear the full cost
- Precautionary → Vic fracking/CSG ban in the constitution
- User-pays → Safeguard Mechanism (>100kt) & rehab bonds
- Intergenerational → Hazelwood pit-lake rehabilitation
- Efficiency → cutting transmission & conversion losses (KK05/06)
- Intragenerational → just transition for Latrobe Valley workers
- Point: name the principle + the stage
- Evidence: a specific Australian example with a figure
- Link: state whether it upholds or undermines — and how fully
- Evaluate? add the "partly, because..." tension
- inter =between gens · intra =within one gen

ECOLOGICAL INTEGRITY RESOURCE-USE EFFICIENCY INTERGENERATIONAL EQUITY INTRAGENERATIONAL EQUITY PRECAUTIONARY USER-PAYS

Ecological integrity Resource-use efficiency Intergenerational equity Intragenerational equity Precautionary User-pays

Every other dot point in this chapter has handed you a piece of an energy source — what it is (KK01–02), what burning it does to the carbon cycle (KK03), how fast we are using it (KK04), how efficiently it converts (KK05), and what the laws of thermodynamics permit (KK06). KK07 is where you assemble those pieces into a judgement. It asks you to run the six sustainability principles like a checklist across the five stages an energy resource passes through on its way from the ground to your power point.

Think of it as a quality-control inspection of an energy supply chain. At every stage — the mine, the well, the refinery, the pipeline, the wall socket — you ask the same six questions. The answers are what separate a coal pathway from a solar one, and they are exactly what examiners want you to apply, compare and evaluate.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation. Why it loses marks: names principles but never applies them to the extraction stage; no Australian example; "bad for the environment" is not ecological integrity; lists three principles vaguely instead of developing one. Mark: identifies a principle (1), but no application and no evidence.

✓ STRONG / MODEL ANSWER

The user-pays principle requires those who extract a resource to bear its full environmental cost. At the extraction stage of Victorian brown coal, the operator of a Latrobe Valley mine must lodge a rehabilitation bond with the State so that the cost of returning the 135 m-deep void to a safe, stable landform is paid by the miner, not the public. This upholds user-pays by internalising the rehabilitation cost into the price of the coal — though it is only partly upheld where the bond (\$289M) is below the estimated full cost (\$743M). Why it scores: defines the principle (1), applies it to the named extraction stage with a specific Australian mechanism + figure (1), and lin

The mine stops. Now someone has to heal the hole.

Why we can't just walk away

A decommissioned energy site isn't neutral ground — it is actively dangerous and degraded. Rehabilitation exists to remove that hazard and hand the land back with a future.

Picture the fictional Yarrabin open-cut, the brown-coal pit that fed the Yarrabin Energy Park for sixty years. When the last shovel-load is burned, it leaves an 18-kilometre-perimeter hole up to 135 metres deep, ringed by near-vertical walls of soft clay and coal. Left alone, that site slumps, erodes, catches fire, and leaches acid and metals into groundwater. Nothing lives there because there is no soil — mining stripped it decades ago.

Australian law forces the operator to fix this. The regulator's benchmark is a single, examinable phrase: the land must be returned to a safe, stable and sustainable landform. Safe = no fire, collapse or toxic exposure risk to people. Stable = the landform won't keep moving, slumping or eroding. Sustainable = it can support a self-maintaining land use (bush, farmland, a lake, recreation) without endless human propping-up.

◆ ANALOGY

The deep-wound analogy Healing a mine is like healing a deep cut. First a surgeon physically closes and stabilises the wound — that's the mechanical work (stitches, setting bone). Only then can the body's living tissue regrow across it — that's the biological work (new skin, blood supply). Skip the stabilising and the living layer just sloughs straight off. Same order, every time.

MECHANICAL · PHYSICAL & ENGINEERING

Two toolkits: mechanical & biological

VCAA splits rehabilitation into exactly two process families. Getting a clean definition of each — and never blurring them — is the core skill this dot point tests.

Non-living, engineering processes that reshape and stabilise the landform and lock away hazards. Bulldozers, excavators, dump trucks and graders do the work.

Processes that use living organisms — plants, microbes and fungi — to rebuild soil, vegetation and a working ecosystem on top of the stabilised landform.

● NOTE

The one-line test: if it's done by a machine to the shape of the land → mechanical. If it's done by something alive, or relies on something alive → biological. Topsoil is the bridge: moving it is mechanical, but what's in it (the living seed bank and microbes) is biological.

LIVE 3D DRAG TO ORBIT · CLICK A LAYER 3D LAYER CROSS-SECTION (WEBGL UNAVAILABLE — USE THE LAYER BUTTONS BELOW). FULL PROFILE — 5 ENGINEERED LAYERS

Inside a rehabilitated landform

◇ ACTIVE RECALL

A rehabilitated site is built up in engineered layers. Peel them back to see where the mechanical work ends and the biological work begins.

• NOTE

The profile, bottom to top: waste-rock backfill rebuilds the bulk landform (mechanical) → a low-permeability clay seal caps the coal so it can't burn or leach (mechanical) → subsoil is deep-ripped and shaped (mechanical) → stored topsoil is respread, reawakening its seed bank and microbes (the bridge) → vegetation roots in and the living ecosystem returns (biological).

Why the order is non-negotiable

A favourite exam discriminator: explaining the sequence. Mechanical work almost always comes first — and there's a clear cause-and-effect reason.

You cannot grow a forest on a 70° wall of slumping clay. The mechanical stage must come first to create a landform that won't move: slopes are battered back to a stable angle, the void is backfilled, drainage is shaped, and the hazardous coal is sealed under clay. Only once the ground is safe and stable does the biological stage make sense — because revegetation needs a surface that stays put long enough for roots to establish.

The crucial hinge between the two is topsoil. Good operators strip and stockpile topsoil before mining begins, then respread it at the end. That single move is the most powerful step in the whole process, because living topsoil already contains the seed bank, microbes and fungi an ecosystem needs — you're not building life from scratch, you're reawakening it.

◆ ANALOGY

Foundations before furniture No one furnishes a house before the slab is poured and the frame is up. Mechanical = the structure (foundation, frame, weatherproofing). Biological = moving in (the living, functioning home). Do them out of order and everything you place inside collapses with the building.

• NOTE

Progressive rehabilitation is the gold standard: rehabilitating worked-out sections while the rest of the site is still being mined, rather than waiting until the very end. It spreads cost and risk and means freshly-stripped topsoil goes straight back down while it's still biologically alive.

LIVE 3D DRAG TO ORBIT 3D REVEGETATION / SUCCESSION SCENE (WEBGL UNAVAILABLE — THE YEAR SLIDER STILL DRIVES THE READOUT). YEAR 0 · BARE RESPREAD TOPSOIL

Bringing the land back to life

◇ ACTIVE RECALL

The biological stage is slow, living and measurable. Watch a stabilised surface move through ecological succession — and see how plants can actually clean the ground.

- **NOTE**

Succession. Pioneer grasses and nitrogen-fixers colonise first, building soil organic matter; shrubs follow; then canopy trees. Returning soil microbes and mycorrhizal fungi are the hidden engine — scientists actually measure the microbial community to judge whether rehabilitation is succeeding.

- ◇ **ACTIVE RECALL**

Direct-seeding and planting native, locally-provenanced species so the cover is self-sustaining and matches surrounding bush. Microbes break down hydrocarbons and other contaminants in soil and water — e.g. microalgae stripping dissolved metals from mine water. Specific plants (like vetiver grass) draw up or stabilise heavy metals through their roots, pulling contamination out of the ground.

Rehabilitate is not restore

This distinction is quietly worth marks. Examiners punish students who treat the two as synonyms.

Returning the land to a safe, stable and sustainable condition fit for a viable land use — which may be different from the original (e.g. a former pit becomes farmland, a recreation lake, or a solar farm). The bar is function and safety, not historical accuracy.

The more ambitious goal of returning the site to its original, pre-disturbance ecosystem — same species, same structure, same function. Far harder, often impossible after deep open-cut mining where the original soil profile is gone forever.

- **NOTE**

Why it matters: after 135 m of open-cut mining the original landform and soil simply no longer exist, so full restoration is usually unrealistic. Most real energy-site projects target rehabilitation to a chosen end land use — which is why the pit-lake option is even on the table.

The Latrobe Valley — Australia's real test case

Yarrabin is fictional; the science and the stakes are not. Victoria is right now attempting the largest mine rehabilitation in its history, on our doorstep in Gippsland.

Three giant brown-coal mines — Hazelwood (closed 2017), Yallourn (closing 2028) and Loy Yang (around 2035) — have powered Victoria since the 1920s. Hazelwood alone is an 18 km-perimeter void up to 135 m deep, with rehabilitation estimated at roughly \$743 million. The Hazelwood mine is bigger than the Melbourne CBD; cleaning it up is the biggest rehabilitation Victoria has ever attempted.

The leading plan for each is a pit lake — slowly filling the void with water to keep the walls stable and the coal submerged so it can't burn. Yallourn's void alone needs an estimated ~630 gigalitres, drawn over decades from the stressed Latrobe/Morwell river system — which is exactly why it's controversial. The mechanical work (battering walls, sealing coal, managing water) and biological work (revegetating the rim and catchment) run together over generations.

The Mineral Resources (Sustainable Development) Act 1990 requires every licensee to rehabilitate to a safe, stable, sustainable landform. The 2026 Financial Assurance Amendment (passed Feb 2026) adds a trailing liabilities scheme so a company can't dodge the cleanup bill.

The Mine Land Rehabilitation Authority oversees Victoria's three declared mines, and Resources Victoria runs the Latrobe Valley Regional Rehabilitation Strategy. Plans go through a full Environment Effects Statement.

Rehabilitation bonds make the operator pre-pay security for the cleanup — a direct application of the user-pays principle. Hazelwood's bond (~\$289 m) sits well below the estimated full cost, a live point of public debate.

How it ties to the sustainability principles

KK08 doesn't live in isolation — examiners love linking rehabilitation back to the principles you met earlier in the course.

Rehabilitating now means we don't dump a dangerous, useless landscape on future generations. An un-rehabilitated pit is a debt passed forward.

Revegetation, bioremediation and reconnecting habitat rebuild functioning ecosystems and biodiversity on degraded land.

Rehabilitation bonds and trailing-liability laws force the company that profited from the energy to fund the cleanup, not taxpayers.

Stabilising walls and sealing coal before failure — and assessing pit-lake water impacts up-front via an EES — acts ahead of proven harm.

Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you what shape your answer must take. Tap each to see what the examiner actually wants for this topic.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

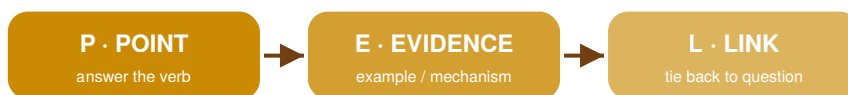
Match your answer to the verb — the fastest marks in the exam.

Six exam traps

⚠ EXAM TRAPS

The predictable ways students drop marks on this dot point. Open each before you sit a SAC.

P-E-L writing trainer



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

P-E-L WRITING

P oint · E vidence/example · L ink. Write a paragraph and watch live which key ideas you've actually hit. Aim to light up every chip.

Worked answer: weak vs strong

Same question, two responses. Read why the examiner's pen moves — then steal the strong version's structure.

• NOTE

Question (4 marks): Describe one mechanical and one biological process used to rehabilitate a former open-cut coal mine, and explain how each contributes to a safe, stable and sustainable landform.

They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before.

A mechanical process is battering down (regrading) the steep highwalls and backfilling the void with overburden, which creates a stable landform that won't slump or collapse — making it safe and stable. A biological process is resspreading stockpiled topsoil and revegetating with native species, whose roots bind the surface against erosion and rebuild a self-sustaining ecosystem — making the landform sustainable.

Practice exam · 22 marks

★ PRACTICE & SELF-MARK

Five VCAA-style questions. Attempt each in your logbook first, then open the model answer and self-mark honestly with the stars. Your score builds in the dock.

One-screen cheat sheet

- Reshape / batter steep walls to stable slope
- Cap / seal coal with clay (stop fire & leaching)
- Deep-rip to relieve compaction
- Contour drainage; build erosion banks
- Plug / cap oil & gas wells
- Respread topsoil (seed bank + microbes)
- Revegetate with native species
- N-fixers & mycorrhizal fungi rebuild soil

- Bioremediation — microbes clean contaminants
- Phytoremediation — plants draw up metals
- Reintroduce fauna; manage weeds
- Goal = safe, stable, sustainable landform
- Mechanical first, then biological
- Topsoil = the bridge between them
- Rehabilitate ≠ restore (original ecosystem)
- Progressive rehab = clean as you mine
- Links: intergen. equity, user-pays (bonds)

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before. "Cleaning it" isn't a named mechanical process — no reshaping, backfill or capping. "Planting trees" is on the right track but vague, and never linked to the goal. "Like before" wrongly implies full restoration; ignores safe/stable/sustainable.

✓ STRONG / MODEL ANSWER

A mechanical process is battering down (regrading) the steep highwalls and backfilling the void with overburden, which creates a stable landform that won't slump or collapse — making it safe and stable. A biological process is resspreading stockpiled topsoil and revegetating with native species, whose roots bind the surface against erosion and rebuild a self-sustaining ecosystem — making the landform sustainable. Names a specific mechanical process AND its function (stability/safety). Names a specific biological process AND its function (sustainability). Explicitly ties both to the marked phrase "safe, stable and sustainable".

Keeping the lights on: base and peak load

THE BALANCING ACT

01 Electricity can't be stockpiled — supply must equal demand every single second

Coal, gas, water and sunshine can be stored. The electricity made from them, mostly, cannot. The amount a grid generates has to match the amount everyone is using at that instant, or the system's frequency drifts away from 50 Hz and protective relays start tripping — that's a blackout.

◆ ANALOGY

Analogy — the tank with no tank Picture a water tower that supplies a town but has no actual storage tank — water flows straight from the pumps to every tap. If everyone showers at 7 am and the pumps don't speed up, pressure collapses. If the pumps over-pump, pipes burst. The electricity grid runs exactly like this: generation in must equal consumption out, moment to moment. "Storage" (batteries, pumped hydro) is the expensive, recently-added exception — and it's the whole reason base/peak load is a live debate.

The 24-hour shape of that demand is the key to the entire dot point. Demand is never flat: it has a floor that is always there, and spikes at predictable times. Engineers split the curve into two ideas — base load and peak load — because the cheapest, cleanest way to meet each is completely different.

BASE VS PEAK LOAD

02 Reading the demand curve

Over any 24 hours, total electricity demand rises and falls in a familiar pattern: low overnight, a morning lift, a dip through the working day, then a sharp evening peak as people get home, cook, heat or cool, and switch everything on at once.

The minimum, continuous level of demand that is present at every hour of the day — the floor of the curve. It comes from things that never switch off: refrigeration, hospitals, water and sewage pumps, street lighting, data centres, and industry running through the night.

On the Yarrabin grid the overnight minimum sits near 2,300 MW — that is the base load.

The maximum demand, reached only briefly — the crest of the curve. It's driven by everyone using power at the same time: the ~6–8 pm cooking/heating/cooling surge, or a 40°C summer afternoon when air-conditioners run flat out.

Yarrabin's evening peak hits about 4,900 MW — more than double the base.

◆ ANALOGY

Analogy — your phone's battery drain Base load is the background drain — the clock, the radios, the always-syncing apps — a small, steady draw you can't switch off. Peak load is opening a graphics-heavy game — a short, intense spike on top of the background. You wouldn't engineer the whole phone around the game, but you do need headroom to handle it when it happens.

◆ KEY INSIGHT

The examinable distinction Base load is not “the biggest source of energy” and peak load is not “renewable energy.” They are parts of the demand curve: base = the always-present floor; peak = the brief maximum. Sources are then judged by how well they can supply each.

MATCHING SOURCES TO LOAD

03 Dispatchability: can you turn it up when you need it?

The single property that decides whether a source suits base or peak load is dispatchability — whether grid operators can raise or lower its output on command. This splits every source into three groups.

Role	Behaviour	Best-suited sources	Why
Base-load	Runs continuously at a steady output; slow to start up or shut down	Black/brown coal, nuclear, large hydro, geothermal, biomass	Cheap to keep running once hot; high capacity factor (~80–90%); inefficient and costly to keep ramping
Peaking	Dispatchable — starts, ramps and stops quickly	Open-cycle gas turbines, hydro (esp. pumped hydro), grid batteries	Fast response fills short spikes; higher cost per unit, so only run during peaks
Variable (intermittent)	Output set by weather/time, not by the operator — must-take	Solar (midday only), wind (when it blows)	Zero fuel cost & emissions, but can't be “turned up” for the evening peak; needs firming

✓ TIP

Capacity ≠ energy Capacity (MW) is how much a generator can produce at full tilt. Energy (MWh) is how much it actually delivers over time. A wind farm rated 100 MW with a 35% capacity factor delivers, on average, only 35 MW of energy — and you can't choose when. Examiners love this gap: high capacity does not mean reliable supply.

GRID DISPATCH SIMULATOR

04 Run the Yarrabin grid for a day

This is the dot point made tangible. The black line is the real demand curve from §02. Your job: build a supply stack beneath it that meets demand at every hour — without big shortfalls (blackouts) or huge surpluses (wasted, curtailed power). Adjust each generator and watch the stack and the gauges respond.

✓ TIP

Try this sequence 1) Hit “coal only” — reliable, but watch the night-time surplus (wasted) and the sluggish evening crest. 2) Hit “solar + wind only” — clean, but the supply collapses at the 6 pm peak: a blackout. 3) Hit “balanced firmed mix” — coal/biomass holds the floor, solar/wind do the daytime, and battery + hydro + gas firm the evening peak. That last move is the modern energy transition in one chart.

THE DUCK CURVE

05 Why more solar makes the evening peak harder

Here's the twist that makes this dot point genuinely current. Rooftop and farm solar don't reduce demand evenly — they slash it only in the middle of the day. What the grid's big generators actually have to supply is net

demand = total demand – solar. As solar grows, that net-demand curve develops a deep midday belly and a steep evening neck. Plotted out, it looks like a duck.

⚠ WATCH OUT

The duck's neck is the problem. The issue isn't midday — there's often too much solar then (prices go negative; output is curtailed). The issue is the evening ramp: as the sun sets, thousands of megawatts must come online in a couple of hours, right as demand peaks. Only fast, dispatchable supply — pumped hydro, batteries, gas — can climb that neck. This is precisely why “just build more solar” doesn't finish the job.

💡 ANALOGY

Analogy — the all-you-can-eat lunch. Solar is a generous friend who shouts everyone a huge free lunch at midday — but vanishes by dinner. Lunch is so big some food is wasted (curtailment). Then at 7 pm everyone's hungry again and your friend is gone, so you have to cook a full meal, fast. The bigger the free lunch, the more dramatic the dinner scramble. Storage is a fridge: it lets you keep some of that free lunch for dinner.

CURRENT & PROJECTED NEEDS

06 From “current” to “projected” — what the VCAA wording means

The dot point asks about current and projected needs. These describe two different grids.

- Base load still leans on ageing coal (e.g. Loy Yang in the Latrobe Valley) and large hydro.
- Peaks met by gas turbines and hydro.
- A fast-growing slice of variable solar and wind, plus the world's highest rooftop-solar uptake.
- Coal stations retire — the base-load floor must be held a new way.
- Far more variable renewables → a deeper duck → more need for firming.
- Storage scales up: pumped hydro (Snowy 2.0), big batteries (Victorian Big Battery, Geelong), home batteries, possibly hydrogen.
- Interconnectors share supply between states; demand management shaves peaks.

💡 KEY INSIGHT

The “firming” idea — write this in evaluation answers. Variable renewables can supply energy cheaply but cannot, alone, guarantee base or peak load on demand. Firming means pairing them with storage and/or fast dispatchable backup so the combined system is reliable. “Renewables + storage” can meet base and peak load; “renewables alone” cannot. That distinction earns marks.

The real Australian context (grounding the Yarrabin fiction): the National Electricity Market (NEM) connects the eastern states; the Australian Energy Market Operator (AEMO) forecasts demand and dispatches generators every 5 minutes to keep supply matched to load. Its Integrated System Plan projects coal exiting and storage + transmission filling the gap.

INDIVIDUAL & SOCIETAL

07 How base and peak needs are met — at two levels

The dot point ends with “at individual and societal levels.” Meeting demand isn't only an engineering job for big operators; households shape the curve too. A strong answer addresses both.

- Generation mix — build base-load + peaking + variable + storage so the stack can meet any hour.

- Dispatch & markets — AEMO/the NEM forecast demand and dispatch the cheapest available generators in real time.
- Storage & firming — pumped hydro and grid batteries store midday surplus for the evening peak.
- Interconnectors — move power between regions to cover local shortfalls.
- Demand response — pay big users to cut load during extreme peaks.
- Rooftop solar + home battery — self-supply by day, store for the evening peak.
- Load shifting — run dishwashers, pool pumps and EV charging off-peak (overnight), flattening the curve.
- Time-of-use tariffs — cheaper off-peak prices nudge this behaviour.
- Energy efficiency — better insulation, efficient appliances and reverse-cycle heating cut both base and peak draw.
- EVs as storage — vehicle-to-grid (V2G) can feed stored charge back during peaks.

✓ TIP

The link between the levels Individual actions aggregate into societal effects. A million homes shifting their dishwasher to 11 pm visibly lowers the evening peak — meaning society needs fewer expensive peaking plants. That's why utilities reward off-peak use: shaving the peak is cheaper than building plant to meet it.

COMMAND WORDS

08 Decode what the question is really asking

◆ COMMAND WORDS

The verb sets the marks. Tap a command word to see what an examiner expects on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

EXAM TRAPS

09 Six ways students lose marks here

◆ NOTE

1 . “Base load = the most energy” No. Base load is the minimum continuous demand (the floor), not the largest amount of energy. Define it as a level of demand, not a quantity of supply.

◆ **NOTE**

2 · “Solar/wind can provide base load” Alone, no — they’re variable, not continuous or dispatchable. Say “solar + storage can help meet base load,” never solar by itself.

◆ **NOTE**

3 · Confusing capacity with energy A 100 MW wind farm rarely delivers 100 MW. Distinguish installed capacity (MW) from delivered energy (MWh) via capacity factor.

◆ **NOTE**

4 · “Batteries generate electricity” They don’t — they store and release it. They shift energy in time (charge midday, discharge at peak). Call them storage, not generation.

◆ **NOTE**

5 · Ignoring “individual and societal” If the question includes both levels, an answer about only the grid (or only households) is capped. Cover both and link them.

◆ **NOTE**

6 · Forgetting the instantaneous match The reason peak load matters is that supply must equal demand right now. Mention the matching/reliability constraint to show you understand why peaking sources exist.

P·E·L TRAINER

10 Write a paragraph that earns the marks



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 **P·E·L WRITING**

VCE extended responses reward structure. Use P·E·L: make a Point, support it with Evidence (a named source, figure or mechanism), then Link back to the question. Type a response below — the meters detect whether each element is present and flag key terminology.

WORKED ANSWER

11 Same question, two responses

Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don’t make as much power. So coal is more reliable but batteries are cleaner.

- **NOTE**

Why it loses marks: no base/peak framing; “batteries are renewable” is wrong (they’re storage, not a source); “don’t make as much power” confuses capacity with role; it’s two separate descriptions, not a true comparison; no dispatchability or capacity-factor language.

Brown coal suits base load: it runs continuously at a high capacity factor (~85%), so it economically holds the always-present demand floor — but it ramps slowly, so it cannot follow short spikes, and it emits heavily.

Grid batteries suit peak load: they are storage, not generation, dispatching stored energy within seconds to cover the brief evening peak, with no direct emissions — but their stored energy is limited, so they cannot sustain base load for long unrecharged.

Overall, the two are complementary rather than competing: coal (or a low-emission base-load source) holds the floor while batteries firm the peak — so a reliable grid needs both roles filled, not one source.

- **NOTE**

Why it scores: explicit base- vs peak-load roles; correct “storage not generation”; capacity factor and dispatchability used precisely; genuine comparison with a synthesising judgement; notes a limitation of each.

PRACTICE QUESTIONS

12 Five graded questions · 20 marks

- ★ **PRACTICE & SELF-MARK**

Attempt each in writing first, then reveal the marking guide and self-mark. Your score feeds the dock in the corner.

KEY TAKEAWAYS

13 If you remember nothing else

- ✦ **RECAP / CHEAT SHEET**

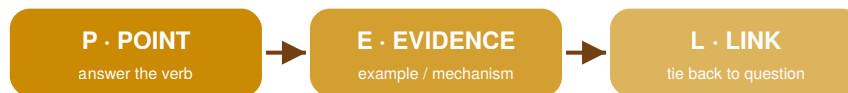
Supply must match demand every second. Electricity is barely storable, so the 24-hour demand curve drives everything. Base load = the continuous minimum demand (the floor); peak load = the brief maximum (the evening/heatwave crest). They’re parts of the demand curve, not types of source. Dispatchability decides the match. Continuous, slow sources (coal, nuclear, geothermal, large hydro) suit base load; fast, dispatchable sources (gas, hydro, batteries) suit peaks. Variable renewables (solar, wind) supply cheap energy but can’t be turned up on demand — so alone they meet neither base nor peak reliably. They need firming (storage + backup). The duck curve: rooftop solar guts midday net demand and steepens the evening ramp — making storage, not more solar, the key to the peak. Capacity (MW) ≠ energy (MWh). Capacity factor bridges them. Batteries store, they don’t generate. Two levels: societal (generation mix, AEMO dispatch, storage, interconnectors, demand response) and individual (rooftop solar + battery, load shifting, efficiency, EVs) — and individual actions aggregate to shave the societal peak.

- ◆ **KEY INSIGHT**

One-line summary A reliable grid fills two jobs — a steady base-load floor and a fast peak-load crest — and the energy transition is really about meeting both as we swap dispatchable fossil fuels for variable renewables backed by storage, at both the grid and household level.

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

1.5/4 Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don't make as much power. So coal is more reliable but batteries are cleaner. Why it loses marks: no base/peak framing; "batteries are renewable" is wrong (they're storage, not a source); "don't make as much power" confuses capacity with role; it's two separate descriptions, not a true comparison; no dispatchability or capacity-factor language.

✓ STRONG / MODEL ANSWER

Brown coal suits base load: it runs continuously at a high capacity factor (~85%), so it economically holds the always-present demand floor — but it ramps slowly, so it cannot follow short spikes, and it emits heavily. Grid batteries suit peak load: they are storage, not generation, dispatching stored energy within seconds to cover the brief evening peak, with no direct emissions — but their stored energy is limited, so they cannot sustain base load for long unrecharged. Overall, the two are complementary rather than competing: coal (or a low-emission base-load source) holds the floor while batteries firm the peak — so a reliable grid needs both roles filled, not one

A Sustainable Energy Future

ORIENTATION

The big idea: a future that runs on less

Every earlier dot point in this chapter asked "where does our energy come from?". This one asks the harder, more examinable question: how do we build an energy system that can keep going indefinitely without wrecking the systems we depend on? The study design gives you three levers to pull — and a good answer pulls all three, not just one.

◆ ANALOGY

The leaky-bucket analogy Picture our energy demand as a bucket you're trying to keep full. The obvious move is to pour in more water — build more power stations. But the bucket is riddled with holes: every conversion wastes heat, every draughty house leaks warmth, every idling engine throws energy away. Plugging a hole is cheaper than fetching more water. Energy you never had to generate is sometimes called a negawatt — the cheapest, cleanest "power station" of all is the one you don't need to build.

So a sustainable energy future isn't only about swapping coal for solar. It's a hierarchy, much like the waste hierarchy (reduce → reuse → recycle) you may know from elsewhere. Work down it in order:

Improve resource efficiency (less raw fuel for the same job) and conversion efficiency (better technology that wastes less at each energy transformation).

Replace finite, high-emission fossil fuels with renewable and low-carbon sources — and electrify the things that currently burn fuel directly (cars, heating).

Cut personal and societal consumption through behaviour, design and demand management — the negawatts that never have to be generated.

✓ TIP

Examiner's lens VCAA rarely asks you to merely list these three. It asks you to evaluate them — to weigh advantages against disadvantages and judge suitability against the sustainability principles (KK07) and base/peak load needs (KK09). Keep the verb "evaluate" in mind the whole way through.

D · REPLACE

Lever 2 — Replacing fossil fuels (interactive)

The Kalduke Valley's Wirndook brown-coal station (our chapter's villain from KK03) is closing. In its place, the Mirran Plains Renewable Energy Zone rises — wind, solar, big batteries and a new transmission line. Drag the year slider to watch the grid transform, and watch what happens to renewables share and emissions.

◆ NOTE

This is real, and it's happening in Victoria The slider tracks Victoria's legislated trajectory: about 42% renewables today, rising to 65% by 2030 and 95% by 2035, with net-zero emissions by 2045. Yallourn closes in 2028, Loy Yang B in 2032 and Loy Yang A by 2035. Grid emissions are modelled to fall from ~41 Mt CO₂-e (2021) to ~3 Mt by 2035.

Replacing isn't just generation. A genuine swap needs four things working together, and a strong exam answer names them:

What you replace	With what	The catch
Coal & gas generation	Wind, solar, hydro, plus a controlling-stake public builder (Victoria's revived SEC, HQ in Morwell)	Output is intermittent — it must be firmed
The gap when the sun sets	Storage: pumped hydro (Snowy 2.0), grid batteries (VIC targets 2.6 GW by 2030, 6.3 GW by 2035)	Storage is finite-duration and costs money & materials
Petrol & gas you burn at home	Electrification: EVs, heat pumps, induction — shift the load onto a clean grid	Only as clean as the grid charging it
The wires themselves	New transmission: VNI West, Western Renewables Link, Marinus Link	Build delays & social licence are the real bottleneck

⚠ WATCH OUT

The trap students fall into "Just build more solar." The Victorian Auditor-General flagged in late 2025 that transmission lines won't be ready before the coal stations shut — so the limiting factor isn't panels, it's poles, wires and storage to firm the supply for base and peak load (KK09). Evaluate the whole system, not one source.

CONCEPT

Lever 1 — Efficiency: resource vs conversion

The dot point names two distinct kinds of efficiency. Examiners love to test whether you can tell them apart, so lock this distinction down.

Using fewer units of the primary resource to deliver the same benefit. Example: a cogeneration plant that captures "waste" heat to warm buildings extracts more useful service from each tonne of gas. Recycling aluminium uses ~5% of the energy of smelting new metal — same product, far less resource.

The fraction of input energy that becomes useful output at each conversion step (KK05). A combined-cycle gas turbine (~55–60%) beats an open-cycle one (~35%). An LED converts far more electricity to light than an incandescent globe. Better tech $\Rightarrow$ smaller waste-heat losses.

◆ ANALOGY

The removalist analogy (why a heat pump seems to break physics — but doesn't) A resistive heater turns 1 unit of electricity into 1 unit of heat (efficiency $\approx$ 100%). A heat pump delivers 3–4 units of heat per unit of electricity (a Coefficient of Performance of 3–4). That looks like $>100\%$ — impossible! But it isn't creating energy; it's a removalist moving existing heat from outside air into your room. Moving boxes is far cheaper than building new ones. The Second Law (KK06) is safe: the pump just relocates low-grade heat, it doesn't manufacture it.

D · EFFICIENCY

Conversion efficiency, made visible (interactive)

✓ TIP

Why transport is the headline win A petrol car is a rolling furnace: only about 20–25% of the fuel's chemical energy reaches the wheels — the rest leaves as heat and noise. An electric vehicle delivers roughly 85–90% of its battery energy to the wheels. Electrifying transport is therefore both a replace move and a giant efficiency move at once.

D • REDUCE

Lever 3 — Reducing personal consumption (interactive)

This is the lever you control. Toggle each measure on the Kalduke Valley demonstration home and watch its yearly energy use, power bill and emissions fall. Notice how cheap behaviour changes stack up against big-ticket upgrades.

◆ NOTE

The policy that makes "reduce" cheaper (2026) From July 2025 the federal Cheaper Home Batteries Program took ~30% off home batteries (tiered down from 1 May 2026); Victoria's Victorian Energy Upgrades scheme subsidises efficient appliances, and new rental thermal-efficiency rules under the 2026–30 climate strategy push insulation into more homes. A Virtual Power Plant (VPP) can even pay you to share stored solar when the grid peaks — turning a household into part of the supply solution.

◆ ANALOGY

The "star ratings" analogy Reducing consumption isn't only sacrifice. Choosing a 5-star fridge over a 2-star one, or air-drying clothes, delivers the same service for less energy — like filling your trolley with the same groceries but paying less at the checkout. The service stays; only the waste shrinks.

SYNTHESIS

Putting it together: evaluating the options

A top-band answer doesn't treat the three levers as a menu to pick from — it shows how they interlock and judges each against the sustainability principles (KK07) and base/peak load reliability (KK09).

Option	Advantage	Disadvantage / limit	Sustainability link
Improve efficiency	Cheapest & fastest; cuts demand, emissions and bills at once	Bounded by the 2nd Law — you can never reach 100%; rebound effect can erode gains	Intergenerational equity; conservation of resources
Replace fossil fuels	Tackles the root cause — finite stocks & CO ₂ ; renewables are now lowest-cost new supply	Intermittent → needs storage & transmission; material & land footprint; transition jobs	Precautionary principle; intergenerational equity
Reduce consumption	Negawatts need no new infrastructure; empowers individuals; eases peak load	Relies on behaviour & equity of access; hard to mandate; limited ceiling alone	Intergenerational & intragenerational equity

✓ TIP

The sentence that earns marks "No single lever is sufficient. Reducing demand and improving efficiency shrink the size of the system we must build, making it feasible to replace the remainder with firm renewables while still meeting base and peak load." If you can write that idea in your own words, you understand KK10.

EXAM SKILL

Command-word decoder

◆ COMMAND WORDS

Same content, different verb, different answer. Tap each card to see exactly what the examiner wants — and how many marks it's usually worth.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

PITFALLS

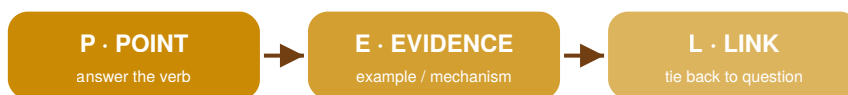
Six exam traps that cost marks

⚠ EXAM TRAPS

These are the recurring ways students lose marks on this dot point. Each is a trap many bright students walk into. A source can be renewable but used unsustainably (e.g. clearing native forest for biomass), or non-renewable yet relatively low-emission in transition (gas as a "firming" fuel). Sustainability is about the whole system and the principles (KK07) — not a label on the fuel. Don't equate the two. Replacing coal needs storage and transmission, not just panels. An answer that ignores firming, base/peak load and the grid itself caps out at mid-band. The dot point names three levers. If the question says "options for a sustainable energy future" and you only write about renewables, you've answered a third of it. Always scan for all three. It has a COP of 3–4, not an efficiency over 100%. It moves heat rather than creating it, so the Second Law isn't violated. Use "coefficient of performance", not "efficiency", for heat pumps. For an "evaluate" command word, a list of options with no weighing of pros/cons and no justified conclusion will not reach the top band. Always finish with a judgement. Efficiency gains can be partly cancelled if cheaper running costs lead people to use more (bigger TVs, longer drives). A sophisticated answer acknowledges rebound as a limit on efficiency-alone strategies.

WRITE & SELF-CHECK

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Strong extended responses follow P·E·L: make a Point, Explain the mechanism, then Link back to the question (sustainability, base/peak, or the judgement). Write your answer below — the trainer lights up the key ideas as you include them. Now compare a model good and bad answer to the same question: Improving efficiency reduces the energy needed for the same service, e.g. a heat pump delivers 3–4 units of heat per unit of electricity (COP 3–4). This cuts demand, emissions and cost simultaneously, and is the fastest, cheapest lever because it needs no new generation — every avoided unit is a "negawatt". However it is bounded by the Second Law and the rebound effect, so it cannot decarbonise alone. Efficiency is therefore highly suitable as a first step that shrinks the system we must then replace and firm.

• NOTE

Point + mechanism + advantage + limitation + justified judgement = top band.

P·E·L WRITING

Efficiency is good because it saves energy and helps the environment. We should use LED lights and turn things off. This is renewable and sustainable so it is a good idea for the future. Everyone should try to be efficient to stop climate change.

• NOTE

No mechanism, no figures, no limitation, no judgement; "renewable" misused; vague. Caps low.

SELF-MARK

VCAA-style practice · 20 marks

★ PRACTICE & SELF-MARK

Five questions building from recall to full evaluation. Attempt each in your logbook first, then reveal the model answer and marking scheme and award yourself stars. Your progress saves automatically. Resource efficiency means using fewer units of the primary resource to deliver the same service (e.g. recycling aluminium uses ~5% of the energy of smelting new metal). Conversion efficiency is the fraction of input energy that becomes useful output at a transformation step (e.g. a combined-cycle gas turbine ~55–60% vs open-cycle ~35%).

- 1 mark — correct distinction (less resource used vs less waste per conversion).
- 1 mark — one valid example for each type.

★ PRACTICE & SELF-MARK

An EV converts ~85–90% of its stored energy to motion, whereas a petrol car wastes ~75–80% as heat — a large conversion-efficiency gain (1). It also shifts the energy source from finite petroleum burned in the car to electricity that can be generated renewably (1). As the grid decarbonises toward 95% renewables, the same EV's lifecycle emissions keep falling, addressing both finite-resource depletion and CO₂ (1).

- 1 mark — efficiency point (EV tank-to-wheel far higher than ICE).
- 1 mark — source/replace point (electricity can be renewable; petrol is finite/high-emission).
- 1 mark — links cleaner grid → falling emissions over time.

★ PRACTICE & SELF-MARK

Similarity: both reduce the household's reliance on fossil-fuelled grid electricity and so cut emissions and bills. Difference (replace): solar+battery is a replacement measure — it generates clean energy and stores it to meet evening peak load, and via a VPP can even support the grid; but it has a high upfront cost and material footprint. Difference (reduce/efficiency): LEDs and the 1°C change are efficiency/reduction measures — extremely cheap "negawatts" with instant payback, but limited in ceiling and partly dependent on behaviour. Judgement: the cheap reductions should come first because they shrink the size (and cost) of the solar+battery system the home then needs.

- 1 mark — a genuine similarity.
- 1 mark — solar+battery framed as replacement + a relevant pro/con (cost, peak-load, VPP).
- 1 mark — LED/behaviour framed as efficiency/reduction + a relevant pro/con.
- 1 mark — a comparative judgement (e.g. cheap reductions first).

★ PRACTICE & SELF-MARK

Advantages: replacing coal/gas tackles the root cause of the problem — depletion of finite stocks and CO₂ emissions — and large-scale wind/solar is now the lowest-cost new generation. It supports intergenerational equity and the precautionary principle. Limitations: renewable output is intermittent, so it must be firmed with storage (VIC targets 2.6 GW by 2030) and backed by new transmission (VNI West, Marinus Link). The Auditor-General warned these lines may lag the coal closures, risking shortfalls in base/peak load; there are also material, land and transition-jobs impacts. Conclusion: replacement is necessary and the dominant long-term strategy, but it is only viable when paired with storage, transmission and demand-side measures — so it should be judged essential but insufficient on its own.

- 1 mark — clear advantage tied to finite resources/emissions or cost.
- 1 mark — link to a sustainability principle (equity / precautionary).
- 1 mark — significant limitation (intermittency, transmission/storage, social/material).
- 1 mark — reference to base/peak load reliability (KK09).
- 1 mark — justified, balanced conclusion (not just "it's good").

★ PRACTICE & SELF-MARK

Support for the statement: Efficiency is bounded by the Second Law — no conversion can be 100% efficient, so waste heat is unavoidable and efficiency alone cannot eliminate energy use. The First Law reminds us energy is conserved, so the service we demand must still be supplied from somewhere — meaning reduction and efficiency lower demand but cannot remove it, leaving a remainder that must be replaced with clean sources. Conversely, replacement alone is harder and costlier the larger the demand, because intermittent renewables need ever more storage and transmission to meet base/peak load. Interaction: reducing consumption and improving efficiency shrink the system, which makes the replacement task smaller, cheaper and faster — they are complementary, not competing. Limitation of the statement: the three are not always equally weighted — replacement does the heavy lifting on emissions, while reduction depends on behaviour and equity of access. Conclusion: the statement is well-justified — the levers are interdependent, and a credible pathway (e.g. Victoria's to net-zero 2045) uses all three, but their balance shifts over time as the grid cleans up.

- 1 mark — correct use of the Second Law (no 100% efficiency / unavoidable waste heat).
- 1 mark — correct use of the First Law (energy conserved → demand must be supplied).
- 1 mark — explains efficiency/reduction shrink the system to be replaced.
- 1 mark — explains why replacement alone is harder at higher demand (storage/transmission/base-load).
- 1 mark — identifies a nuance/limitation (unequal weighting, behaviour, equity).
- 1 mark — justified overall judgement on the statement.

REVISION

One-screen cheat sheet

- Resource: less raw fuel per service (cogeneration, recycling)
- Conversion: less waste per step (LED, CCGT, heat pump COP 3–4)
- Cheapest, fastest lever = "negawatts"
- Limit: 2nd Law + rebound effect
- Renewables + storage + transmission + electrification
- VIC: 95% renewables 2035, net-zero 2045
- Coal out: Yallourn '28, Loy Yang B '32, A '35
- Limit: intermittency → must be firmed
- Insulate, draught-proof, star-rated appliances
- Behaviour: thermostat, active transport, switch-off
- Eases peak load; no new infrastructure
- Limit: behaviour & equity of access
- "Evaluate" = pros + cons + judgement
- Link to sustainability principles (KK07)
- Link to base/peak load reliability (KK09)
- All three levers interact — none sufficient alone

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Incandescent globe 5 % useful

✓ STRONG / MODEL ANSWER

HIGH-BAND RESPONSE Improving efficiency reduces the energy needed for the same service, e.g. a heat pump delivers 3–4 units of heat per unit of electricity (COP 3–4). This cuts demand, emissions and cost simultaneously, and is the fastest, cheapest lever because it needs no new generation — every avoided unit is a "negawatt". However it is bounded by the Second Law and the rebound effect, so it cannot decarbonise alone. Efficiency is therefore highly suitable as a first step that shrinks the system we must then replace and firm. Point + mechanism + advantage + limitation + justified judgement = top band.

✗ WEAK ANSWER

LOW-BAND RESPONSE Efficiency is good because it saves energy and helps the environment. We should use LED lights and turn things off. This is renewable and sustainable so it is a good idea for the future. Everyone should try to be efficient to stop climate change. No mechanism, no figures, no limitation, no judgement; "renewable" misused; vague. Caps low.

✓ STRONG / MODEL ANSWER

Resource efficiency means using fewer units of the primary resource to deliver the same service (e.g. recycling aluminium uses ~5% of the energy of smelting new metal). Conversion efficiency is the fraction of input energy that becomes useful output at a transformation step (e.g. a combined-cycle gas turbine ~55–60% vs open-cycle ~35%). Marking scheme 1 mark — correct distinction (less resource used vs less waste per conversion). 1 mark — one valid example for each type.

Responsible decision-making for a sustainable energy future

The previous ten dot points asked what our energy options are and how good each one is. This final dot point asks something harder: how should the decision actually be made — and by whom? Real energy decisions are not solved by a single fact. They emerge from four forces that pull on each other: stakeholder values, regulatory frameworks, scientific data and new technologies.

THE BIG IDEA

"Responsible" is about the process, not just the outcome

A common mistake is to think this dot point wants you to pick a winner — "wind is best, so build wind." It doesn't. The study design asks you to explain the interconnections and tensions between the factors that influence responsible decision-making. In other words: examine the machinery of the decision, not just the answer it spits out.

A responsible decision is one that is transparent, informed by the best available evidence, considers all affected stakeholders (including those who cannot speak for themselves — future generations and ecosystems), weighs the trade-offs openly, and can be justified against the sustainability principles. The "cheapest" option or "whatever the loudest group wants" is not automatically the responsible one.

◆ ANALOGY

Analogy · the mixing desk Think of a sound engineer at a mixing desk. There's no single "correct" position for every slider — push the bass too high and the vocals vanish; chase clarity and you lose warmth. A good engineer doesn't pretend there's one right answer; they balance competing demands on purpose, and can explain every choice. Energy decision-making is the same desk: cost, reliability, emissions, jobs, equity and amenity are all sliders, and responsible decision-makers tune them transparently rather than slamming one to maximum.

• NOTE

Watch the verb. Exam questions on this dot point almost always use analyse, evaluate, discuss or justify — never "list". You are being asked to reason about how factors interact and trade off, then reach a weighed judgement. Description alone caps you at the lowest marks.

THE FOUR FACTORS

What pushes on every energy decision

The study design names four factors. Learn them as a set — almost every KK11 question is built from two or more of them.

What different people and groups care about — shaped by their value system (anthropocentric, biocentric, ecocentric, technocentric) and their priorities: jobs, bills, reliability, environment, culture, amenity, equity, profit.

The rules of the game — laws, legislated targets, planning and environmental approvals, permits, licences and treaties. They can drive change (mandates, subsidies) or slow it (approval delays, disputes).

The evidence base — emissions, resource availability, efficiency figures, supply-and-demand modelling, climate projections and environmental impact studies. Often uncertain, incomplete or contested.

Emerging options that change what's possible — offshore wind, grid-scale batteries, pumped hydro, green hydrogen, smart grids, EVs. Each carries a gap between promise and proven maturity / cost.

Factor	In the Tarra Cove / Kalduke decision
Stakeholder values	Coal workers want jobs; fishers fear losing grounds; Traditional Owners want a say over sea country and benefit-sharing; households want low bills + reliable power; environmental groups want fast decarbonisation and protected whales/seabirds.
Regulatory frameworks	Victoria's legislated renewable-energy and net-zero targets push the transition; federal environmental approval (EPBC) and offshore-electricity licensing, plus planning permits and cultural-heritage law, gate how fast it can happen.
Scientific data	Wind-resource and bathymetry surveys, whale-migration and seabird studies, grid-stability modelling, and lifecycle-emissions figures — much of it new, with real uncertainty bands.
New technologies	Floating/fixed offshore turbines, big batteries, pumped hydro in mine voids, and green hydrogen — some mature, some still scaling, each with cost and reliability question marks.

STAKEHOLDERS & VALUES

Same data, different verdicts — because values differ

Stakeholders rarely disagree because one side "has the facts wrong". They disagree because they value different things. Identifying each group's value system and priority is the single highest-value skill in this dot point — it turns a vague "people argued" into a precise tension you can analyse.

Stakeholder	Dominant value system	Top priority
Wirndook workers & union	Anthropocentric	Secure jobs & a "just transition"
Tarra Cove fishing co-op	Anthropocentric	Keep access to fishing grounds & livelihoods
Traditional Owners	Ecocentric / cultural	Caring for sea Country, consent & benefit-sharing
Households / consumers	Anthropocentric	Affordable, reliable electricity
Environmental group	Ecocentric / biocentric	Rapid emissions cuts and local wildlife
Renewable developer	Technocentric	Profit & planning certainty
State government	Anthropocentric (mixed)	Meet legislated targets, cost, reliability, votes
Future generations	— (cannot speak)	A liveable climate (intergenerational equity)

• NOTE

Pro move. Whenever a question mentions a stakeholder, name their value system and their priority in the same breath. "The fishing co-op (anthropocentric — livelihoods) opposes the turbines because..." reads like a Year 12 answer; "the fishermen don't like it" does not.

There is no objectively correct setting. Drag the sliders to weight what you think matters most, and watch the "decision" tilt toward a different pathway. The point: the outcome depends on whose values you privilege — which is exactly why responsible decisions must make their weighting transparent and justified.

◆ NOTE

CURRENT VERDICT Balanced transition Your weighting spreads fairly evenly — a phased build with community and environmental safeguards.

INTERCONNECTIONS

The factors feed each other — it's a web, not a line

"Interconnection" means the factors reinforce, enable or depend on one another. A change in one ripples through the rest. Examiners love this because it tests whether you see the system, not four isolated boxes.

This factor...	...connects to...	Worked example at Tarra Cove
Scientific data	Regulation	Climate projections and emissions data justify the legislated net-zero target that drives the whole transition.
Regulation	New technology	Targets + subsidies make the offshore wind and battery projects financially viable enough to build.
New technology	Scientific data	Building the wind farm generates new monitoring data on whales and seabirds, feeding back into the evidence base.
Stakeholder values	Regulation	Community pressure and votes shape which targets and approval conditions governments actually legislate.
New technology	Stakeholder values	Affordable batteries shift what households expect ("I can store my solar"), changing what they value and demand.

◆ ANALOGY

Analogy · the spider's web Touch any strand of a spider's web and the whole structure trembles. Pull the "scientific data" strand (new whale study) and it tugs on regulation (approval conditions), technology (turbine placement) and values (community trust) all at once. Responsible decision-makers feel the whole web move; weak analysis treats each strand as if it were a separate string.

TENSIONS · THE HEART OF THE DOT POINT

Where the factors pull against each other

A tension is a genuine conflict — satisfying one factor means compromising another. There is no setting that keeps everyone fully happy. The skill being assessed is your ability to name the two things in conflict and explain why they cannot both be maximised.

◆ KEY INSIGHT

The "energy trilemma". Many tensions collapse into three goals that can't all be maxed at once: affordable, reliable, clean. Push hard on "clean + reliable" (lots of storage + firming) and cost rises; push on "affordable + reliable" and you lean on fossil fuels. Naming the trilemma is a fast way to frame a tension question.

CONSEQUENCES OVER TIME

Three pathways, three 2045s

Decisions aren't one-off — they unfold over decades, and different weightings of the four factors lead to genuinely different futures for Tarra Cove. Switch between the three pathways below to see how the trade-offs play out. Notice that each pathway is "responsible" to someone — the question is always responsible according to which values and which principles?

Pathway	Whose values it serves	Sustainability trade-off
Delay & protect	Workers, cost-focused households	Protects jobs & bills now; fails intergenerational equity & the legislated target.
Rush the build	Climate-focused groups, developers	Cuts emissions fast; risks ecological integrity & intragenerational equity through thin consultation.
Balanced transition	Spreads benefit across most stakeholders	Upholds the most principles at once; costs more time & coordination — slower decarbonisation than "rush".

THE DECISION-MAKER'S TOOLKIT

How to turn a mess of factors into a defensible choice

"Responsible" isn't a vibe — it's a set of tools and principles you can name. Reach for these whenever a question asks you to evaluate or justify a decision. They're the difference between an opinion and an argument.

Precautionary principle, inter - & intra generational equity, resource-use efficiency, ecological integrity, user-pays. Test every option against these.

Anthropocentric, biocentric, ecocentric, technocentric. Naming a stakeholder's lens explains why they prioritise what they do.

Cost–benefit analysis weighs total benefits vs costs (incl. environmental); qualitative risk analysis ranks likelihood × consequence. Both make trade-offs explicit.

Stakeholder engagement, environmental effects statements and benefit-sharing give the decision a social licence and address equity tensions.

◆ NOTE

The four marks of a responsible decision Evidence-informed — uses the best available scientific data, and acknowledges its uncertainty rather than hiding it. Inclusive — considers all stakeholders, including ecosystems and future generations who have no voice. Transparent about trade-offs — states openly what is being sacrificed and why, instead of pretending there's a free lunch. Principle-justified — can be defended against the sustainability principles, not just "it was cheapest" or "the loudest group won".

COMMAND-WORD DECODER

Do exactly what the verb asks

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

GOOD VS WEAK · ANNOTATED

The same question, two answers

Discuss one tension between scientific data and stakeholder values in the proposal to build an offshore wind farm off Tarra Cove.

The scientists did studies and said the wind farm is safe for the birds. But the fishermen still don't want it because they don't like it. So there is a disagreement between them. The government should look at the science and decide.

Seabird surveys indicate a low collision risk (scientific data), which on its own would support approval. However, the Tarra Cove fishing co-op (anthropocentric — livelihoods) opposes the project because their concern is about lost access to fishing grounds and amenity, not bird safety. This is a tension because favourable data cannot resolve a conflict rooted in values — the co-op's priority is economic security, which the science doesn't address. A responsible decision therefore can't simply "follow the data"; it must also weigh intragenerational equity and pursue consultation or benefit-sharing to address the co-op's loss.

◆ KEY INSIGHT

The pattern that scores: name the data → name the stakeholder (value system + priority) → explain why they genuinely conflict → link to a sustainability principle or process. Four moves, four marks.

WRITING TRAINER · P·E·L

Build a body paragraph that examiners reward



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

P-E-L WRITING

Strong KK11 paragraphs follow Point → Evidence → Link. Write a paragraph below answering: "Explain one tension between regulatory frameworks and new technologies in the Tarra Cove transition." The trainer scans live for the moves that earn marks.

• NOTE

Model move: Point — approval frameworks lag fast-moving technology. Evidence — floating turbines and green hydrogen are ready before the licensing rules are, so strict approvals can delay projects needed to meet the legislated target. Link — this pits the precautionary principle against the need for timely action, so responsible decisions must streamline approvals without abandoning environmental safeguards.

PRACTICE EXAM · 20 MARKS

Five graded questions — write, reveal, self-mark

★ PRACTICE & SELF-MARK

Attempt each in your logbook (or the box), then reveal the marking scheme and award yourself marks. Your running total tracks in the dock, bottom-right, and is saved on this device. Distinguish between an interconnection and a tension among the factors influencing a sustainable energy decision. Support each with one example.

- 1 mark — interconnection = factors reinforce / depend on / enable each other; e.g. scientific data justifies a regulated target.
- 1 mark — tension = factors conflict so both cannot be fully satisfied; e.g. fast decarbonisation (climate values) vs protecting coal jobs.

★ PRACTICE & SELF-MARK

Explain how the priorities of two different stakeholders in the Tarra Cove decision conflict, naming each stakeholder's value system.

- 1 mark — stakeholder A identified with value system + priority (e.g. environmental group, ecocentric, rapid emissions cuts).
- 1 mark — stakeholder B identified with value system + priority (e.g. Wirndook workers, anthropocentric, secure jobs).
- 1 mark — explains why the priorities conflict (fast coal closure cuts emissions but removes the jobs the workers depend on).

★ PRACTICE & SELF-MARK

Analyse how scientific data and regulatory frameworks are interconnected in driving the shift away from brown coal, and identify one tension this can create.

- 1 mark — climate/emissions data provides the evidence base for action.
- 1 mark — this data justifies / shapes regulation (e.g. legislated net-zero & renewable targets).
- 1 mark — clear statement of the interconnection (data → underpins → regulation; they reinforce each other).

- 1 mark — a resulting tension (e.g. targets demand fast closure, but data on jobs/grid reliability urges caution — speed vs precaution).

★ PRACTICE & SELF-MARK

Discuss the tensions a decision-maker must balance when deciding how quickly to approve the offshore wind farm, referring to at least three of the four factors.

- Regulation: legislated targets push for speed; approval/EPBC processes require time.
- New technology: turbines are ready, but rules & long-term data lag — precaution vs progress.
- Scientific data: whale/seabird studies may be incomplete; precautionary principle counsels caution.
- Stakeholder values: fishers & Traditional Owners need genuine consultation, which takes time; climate groups want speed.
- Trilemma / equity: speed serves intergenerational equity & the target; haste risks ecological integrity & intragenerational equity (local community).
- Discussion quality: presents both "go fast" and "go slow" sides rather than one.

★ PRACTICE & SELF-MARK

Evaluate whether the "balanced transition" pathway — phased coal closure with staged renewables, full consultation and a valley jobs program — is the most responsible decision for Tarra Cove's energy future. Refer to all four factors and at least two sustainability principles, and reach a justified conclusion.

- 1 mark — addresses stakeholder values (spreads benefit; consultation gives social licence).
- 1 mark — addresses regulation (meets legislated targets while satisfying approval requirements).
- 1 mark — addresses scientific data (staged build lets monitoring data refine each phase — adaptive, precautionary).
- 1 mark — addresses new technology (phasing lets maturing tech — storage, hydrogen — be added as it proves out).
- 1 mark — applies ≥2 sustainability principles (e.g. intergenerational equity, ecological integrity, intragenerational equity, precautionary principle).
- 1 mark — clear, justified judgement that weighs the main trade-off (e.g. responsible because it upholds the most principles, accepting slower decarbonisation and higher coordination cost than "rush").

★ PRACTICE & SELF-MARK

Top answers note that "most responsible" still involves a real trade-off — there is no costless option — and defend the chosen balance explicitly.

ONE-SCREEN RECALL

KK11 cheat sheet

- Stakeholder values & priorities
- Regulatory frameworks
- Scientific data
- New technologies

♦ RECAP / CHEAT SHEET

Interconnect = factors reinforce/depend on each other. Tension = factors conflict; both can't be maxed. Many questions want both. Always pair a stakeholder with their value system (anthro-/bio-/eco-/techno-centric) and their priority. Remember the voiceless: future generations & ecosystems. Affordable · reliable · clean — you can't max all three at once. A fast frame for most tension questions. Precautionary · inter- & intra-generational equity · ecological integrity · resource-use efficiency · user-pays. Evidence-informed · inclusive of all stakeholders · transparent about trade-offs · justified against principles.

Responsible decision-making for a sustainable energy future

The previous ten dot points asked what our energy options are and how good each one is. This final dot point asks something harder: how should the decision actually be made — and by whom? Real energy decisions are not solved by a single fact. They emerge from four forces that pull on each other: stakeholder values, regulatory frameworks, scientific data and new technologies.

◆ KEY INSIGHT

Set the scene — Tarra Cove & the Kalduke Valley. The brown-coal Wirndook Power Station in the Kalduke Valley is scheduled to close in 2032, ending ~500 local jobs. On the table: a large offshore wind farm in Bass Strait off Tarra Cove, a Kalduke Valley Renewable Energy Zone (solar + grid batteries + pumped hydro in old mine voids), and possible green hydrogen. Everyone agrees the future must be sustainable. Nobody agrees on how to get there. We'll use this one decision all the way through.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The scientists did studies and said the wind farm is safe for the birds. But the fishermen still don't want it because they don't like it. So there is a disagreement between them. The government should look at the science and decide. No value system named · "don't like it" isn't a reason · treats data as decisive · no real tension explained · no principle.

✓ STRONG / MODEL ANSWER

Seabird surveys indicate a low collision risk (scientific data), which on its own would support approval. However, the Tarra Cove fishing co-op (anthropocentric — livelihoods) opposes the project because their concern is about lost access to fishing grounds and amenity, not bird safety. This is a tension because favourable data cannot resolve a conflict rooted in values — the co-op's priority is economic security, which the science doesn't address. A responsible decision therefore can't simply "follow the data"; it must also weigh intragenerational equity and pursue consultation or benefit-sharing to address the co-op's loss. Names the data · names stakeholder + value sy

